

MECHANICAL GENERAL NOTES	
1.	INTENT OF THESE NOTES AND MECHANICAL NOTES ON DRAWINGS IS TO CLARIFY THE SCOPE OF WORK AND ALERT CONTRACTOR OF EXISTING CONDITIONS. CONTRACTOR TO VISIT SITE AND VERIFY ALL CLEARANCES BEFORE FABRICATION OF DUCTWORK AND PROVIDE ADDITIONAL OFFSET AND/OR CHANGES IN DUCT SIZES TO MEET FIELD CONDITIONS AND COORDINATE WITH ELECTRICAL, PLUMBING AND FIRE PROTECTION SUBCONTRACTOR BEFORE ANY CONSTRUCTION WORK.
2.	ALL SUPPLY AND RETURN DUCTWORK SHALL BE EXTERNALLY INSULATED SHEET METAL CONSTRUCTION. FLEXIBLE DUCTS EITHER: SPIRAL ROUND SPRING STEEL WITH FLAMEPROOF VINYL SHEATHING, OR CORRUGATED ALUMINUM, COMPLY WITH UL 181. PROVIDE 1" THICK CONTINUOUS FLEXIBLE FIBERGLASS SHEATH WITH VINYL VAPOR BARRIER JACKET. ALL EXHAUST DUCTWORK SHALL BE UNINSULATED SHEET METAL CONSTRUCTION IN ACCORDANCE WITH LATEST SMACNA STANDARDS. ALL JOINTS SHALL BE SECURELY TAPED WITH 3" WIDE GLASS FABRIC TAPE WITH FOSTER 30/35 MASTIC OR EQUAL. SUPPLY & RETURN AIR DUCTBOARD (DUCT MATERIAL) CAN BE USED IF INSTALLED PER THE 2023 MECHANICAL/ENERGY CODES & SMACNA STANDARDS.
3.	DUCT SHALL BE SECURELY SUPPORTED, HUNG OR SUSPENDED IN ACCORDANCE WITH SECTION 603 OF THE FLORIDA BUILDING CODE - MECHANICAL 2023. EQUIPMENT SHALL BE SUPPORTED FROM STRUCTURAL MEMBERS AS REQUIRED TO PROVIDE A VIBRATION-FREE RIGID INSTALLATION.
4.	EXHAUST FANS AND PLUMBING VENTS SHALL BE LOCATED A MINIMUM OF 10 FEET FROM AIR INTAKES.
5.	PROVIDE DUCT MOUNTED SMOKE DETECTOR IN THE SUPPLY AIR DUCTWORK WIRED TO SHUT DOWN THE SUPPLY FAN, ONLY FOR FANS WITH MORE THAN 2,000 CFM.
6.	MANUFACTURERS NAMES AND MODEL NUMBERS LISTED IN THE SPECIFICATIONS OR ON THE SCHEDULES ARE FOR THE PURPOSE OF ESTABLISHING A QUALITY OF MANUFACTURER OR A SPECIFIC DESIGN CONFIGURATION. EQUAL PRODUCTS, AS APPROVED BY THE ARCHITECT/ENGINEER WILL BE ACCEPTABLE FROM OTHER MANUFACTURERS.
7.	DUCTWORK LOCATIONS AND CLEARANCES SHALL BE COORDINATED WITH ARCHITECTURAL, PLUMBING, ELECTRICAL AND SYSTEMS TRADES TO FIT THE SPACES PROVIDED. DUCTWORK CROSS SECTIONAL AREAS MAY NOT BE REDUCED.
8.	IN GENERAL, PLANS AND DIAGRAMS ARE SCHEMATIC ONLY AND SHOULD NOT BE SCALED. HVAC CONTRACTOR IS RESPONSIBLE FOR ANY ADDED ELECTRICAL COSTS WHICH MAY RESULT FROM SUBSTITUTED EQUIPMENT.
9.	DUCTWORK INSULATION SHALL BE 1-1/2" FOIL FACED FIBERGLASS DUCT WRAP.
10.	COORDINATE AIR DEVICE LOCATIONS WITH LIGHTING FIXTURES. COORDINATE DIFFUSER AND REGISTER LOCATIONS WITH ELECTRICAL LIGHTING LOAD AND ARCHITECTURAL REFLECTED CEILING PLANS.
11.	TURNING VANES SHALL BE PROVIDED IN ALL SUPPLY DUCT RECTANGULAR ELBOWS WITH ANGLES BETWEEN 15 DEGREES AND LESS THAN 90 DEGREES PER FIG. 2-5 OF THE SMACNA MANUAL.
12.	DUCT DIMENSIONS SHOWN ON DRAWINGS ARE CLEAR INSIDE DIMENSIONS. DUCTWORK, DIFFUSERS, REGISTERS, GRILLES, AND OTHER ITEMS OF THE AIR HANDLING SYSTEM SHALL NOT BE SUPPORTED BY THE CEILING OR CEILING SUSPENSION SYSTEM.
13.	ALL SUPPLY AIR DIFFUSERS SHALL BE 4-WAY THROW UNLESS OTHERWISE NOTED.
14.	UNLESS OTHERWISE NOTED, INSTALL DUCTWORK AS HIGH AS POSSIBLE, TIGHT TO BOTTOM OF STRUCTURE. COORDINATE DUCT ELEVATION WITH RAIN LEADERS, WATER PIPING, SANITARY DRAINS AND MAJOR ELECTRICAL CONDUITS.
15.	CONTRACTOR SHALL PROVIDE ALL SUPPLEMENTARY STEEL REQUIRED TO SUSPEND MECHANICAL EQUIPMENT AND MATERIALS. INSTALLATION OF THE EQUIPMENT SHALL COMPLY WITH THE MANUFACTURER'S SPECIFICATIONS AND CLEARANCE REQUIREMENTS FOR SERVICING OF EQUIPMENT.
16.	VERIFY VOLTAGE WITH ELECTRICAL BEFORE ORDERING EQUIPMENT. CONNECTION TO EQUIPMENT SHALL BE VERIFIED WITH MANUFACTURER'S CERTIFIED DRAWINGS. TRANSITIONS TO ALL EQUIPMENT SHALL BE VERIFIED AND PROVIDED FOR EQUIPMENT FURNISHED.
17.	PROVIDE A TRAP IN ALL CONDENSATE PIPING LOCATED AT THE ROOFTOP UNIT. SLOPE CONDENSATE LINES 1/8" PER FOOT. CONDENSATE LINES SHALL BE PVC SCH. 40. ALL CONDENSATE DRAIN PIPING ON ROOF SHALL BE PROPERLY SUPPORTED. SEE "CONDENSATE PIPE SUPPORT ON ROOF" DETAIL.
18.	ALL MECHANICAL WORK SHALL MEET ALL THE REQUIREMENTS OF THE "FLORIDA 2023 BUILDING CODE - MECHANICAL".
19.	GARANTEE: FOR ONE YEAR AFTER DATE OF ACCEPTANCE BY THE OWNER, ALL EQUIPMENT, MATERIALS AND WORKMANSHIP TO BE FREE FROM DEFECT
20.	DO NOT CUT STRUCTURAL MEMBERS WITHOUT PRIOR WRITTEN APPROVAL OF THE STRUCTURAL ENGINEER OR ARCHITECT. ARRANGE FOR REPAIRS REQUIRED TO RESTORE OTHER WORK, BECAUSE OF DAMAGE CAUSED AS A RESULT OF MECHANICAL INSTALLATIONS.
21.	FLEXIBLE AND RIGID ROUND DUCT TAKE-OFFS FOR DIFFUSERS SHALL BE THE SAME SIZE AS DIFFUSER NECK. FLEXIBLE DUCT SHALL BE THERMAFLEX TYPE M-KA OR EQUAL, AND BE UL LISTED AND COMPLY WITH NFPA STANDARD NO.90A.
22.	ALL WALL MOUNTED THERMOSTATS SHALL BE INSTALLED AT AN ELEVATION OF 54" ABOVE FINISHED FLOOR TO THE TOP, EXCEPT FOR HANDICAP ACCESSIBLE UNITS TO BE NO MORE THAN 48" ABOVE FINISHED FLOOR AND UNLESS OTHERWISE NOTED ON DRAWINGS. LOCATION OF THE WALL MOUNTED THERMOSTAT SHALL BE COORDINATED WITH OTHER TRADES FOR A NEAT APPEARANCE. FINAL LOCATION OF THERMOSTAT SHALL BE SUBJECT TO THE APPROVAL OF THE ENGINEER OR HIS REPRESENTATIVE IN THE FIELD.
23.	ALL UNITS SHALL BE BALANCED TO WITHIN 10 PERCENT OF THE DESIGN AIR QUANTITY. BALANCE DIFFUSERS AND REGISTERS TO WITHIN 10 PERCENT OF QUANTITIES SHOWN ON DRAWINGS OR REFER TO SPECIFICATIONS THIS SHEET.
24.	PROVIDE FLEXIBLE NEOPRENE DUCT CONNECTORS ON THE DISCHARGE AND ENTERING SIDES OF ALL VIBRATING EQUIPMENT TO WHICH DUCTWORK IS ATTACHED.
25.	DRYER VENTS IF ANY WITH A TOTAL DEVELOPED LENGTH WHICH EXCEEDS TWENTY-FIVE (25'-0") OR MORE SHALL HAVE A U.L. LISTED DRYER BOOSTER FAN EXCEPT WHERE DRYERS (AS SELECTED BY THE OWNER) HAVE A MANUFACTURER'S LISTING (PUBLISHED) ALLOWING VENT RUNS IN EXCESS OF TWENTY-FIVE (25) EQUIVALENT FEET.
26.	CONTRACTOR SHALL PROVIDE A COPY OF A TEST AND BALANCE REPORT. AIR BALANCING SHALL BE PERFORMED BY AN INDEPENDENT TESTING AND BALANCING CONTRACTOR WHICH SHALL BE QUALIFIED AND CERTIFIED BY EITHER ASSOCIATED AIR BALANCE COUNCIL (AABC) OR NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB). THIS REPORT MUST BE REVIEWED AND APPROVED BY THE ENGINEER PRIOR TO THE FINAL INSPECTION. (OR REFER TO SPECIFICATIONS THIS SHEET)
27.	THE GENERAL CONTRACTOR IS RESPONSIBLE FOR ALL TRADES INSTALLATION SCHEDULES. FIXED WORK SUCH AS DUCTWORK AND PLUMBING SHALL BE INSTALLED PRIOR TO ANY TRADE WORK THAT CAN BE EASILY RELOCATED OR OFFSET SUCH AS ELECTRICAL CONDUITS, SMALL WATER LINES ETC.
28.	REVIEW DRAWINGS AND PROVIDE ALL WORK FOR A COMPLETE AND OPERABLE SYSTEM, INCLUDING ALL INCIDENTALS REQUIRED BY CODE AGENCIES AND LOCAL GOVERNING BODIES. ANY DISCREPANCIES NOT BROUGHT TO THE ATTENTION OF THE ARCHITECT IN WRITING PRIOR TO BIDS WILL BE CONSIDERED CORRECT BY THE CONTRACTOR IN HIS BID TO THE SATISFACTION OF THE ARCHITECT AND AS THE ARCHITECT DIRECTS.
29.	ERRORS OR OMISSIONS: WHEN THE INTENT OF ARCHITECT/ENGINEER WITH REGARD TO ANY DETAIL IS NOT CLEAR, OR IS CAPABLE OF MORE THAN ONE INTERPRETATION, SUCH MATTERS WILL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT/ENGINEER IN WRITING BEFORE THE SUBMISSION OF BIDS, AND THE ARCHITECT/ENGINEER SHALL MAKE CORRECTION OR EXPLANATION IN WRITING. OTHERWISE, NO EXTRA CHARGE WILL BE ALLOWED FOR THE WORK OR MATERIAL WHICH THE ARCHITECT/ENGINEER WILL REQUIRE, PROVIDED THAT IT COMES WITHIN A REASONABLE INTERPRETATION OF THE DRAWINGS AND SPECIFICATIONS.

ALL WORK TO COMPLY WITH THE 2023
FLORIDA MECHANICAL &
ENERGY CODES – 8TH ED

SYMBOL	DESCRIPTION
	NEW DUCTWORK
	FLEXIBLE DUCT
	DUCTWORK UP THRU PLAN
	DUCTWORK TRANSITION
	MANUAL AIR VOLUME CONTROL DAMPER
	SUPPLY AIR DEVICE
	RETURN AIR DEVICE
	EXHAUST AIR DEVICE
	WALL MOUNTED THERMOSTAT
	TEMPERATURE SENSOR IN RETURN DUCT AIR STREAM OR AS SHOWN ON DRAWINGS
	OUTSIDE AIR MOTORIZED DAMPER, INTERLOCK W/ AHU
	DEVICE TYPE "A" - 300 CFM - 8"Ø NECK SIZE
	REFRIGERATION PIPING
	CONDENSATE PIPING
	FIRE DAMPER, SMOKE DAMPER WITH ACCESS DOOR

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[M] [P] DESIGN ENGINEERING, INC. ARCHITECTURAL ELECTRICAL PLUMBING FIRE PROTECTION 6000 W. US Highway 90 #200 Wilder Park, FL 32792 (407) 716-8238 Certificate #280116 www.floridaplumbing.org © Copyright 2024 EUT Engineering, Inc.																													
•  MEDICAL OFFICE  • SOUTHERN RAILWAY BLDG #9 OAKLAND, FL 34787		PROJECT																											
MECHANICAL SPECIFICATIONS, ABBREVIATIONS, LEGEND AND NOTES		SECTION																											
N. A. S. AHMED FLORIDA P.E.# 56095 <table border="1" style="width: 100%; border-collapse: collapse; font-size: x-small;"> <thead> <tr> <th colspan="2">REVISONS:</th> <th></th> </tr> <tr> <th>#</th> <th>DATE</th> <th>DESCRIPTION</th> </tr> </thead> <tbody> <tr> <td style="background-color: #d3d3d3;">A</td> <td>02/16/26</td> <td>SUBMIT</td> </tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> </tbody> </table> DRAWN BY: AFS CHECKED BY: AFS DATE: JANUARY 30, 2026 SHEET: M000 Elite Universal Job No. 90210			REVISONS:			#	DATE	DESCRIPTION	A	02/16/26	SUBMIT																		
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VERTICAL AHU / SPLIT SYSTEM A/C UNIT SCHEDULE - R-454B

SELECTION BASED ON: TRANE OR EQUAL

MARK	AREA SERVED	NOMINAL TONS	AIR HANDLING UNIT - SINGLE POINT CONNECTION								MCA	MOCP	WEIGHT (LBS)	COOLING CAPACITY		CONDENSING UNIT				EQUAL TO		
			FAN					ELEC. HEATER								V / Ø	MCA	MOCP	WEIGHT (LBS)	SEER2 (EER)	MFR	MODEL
			CFM	O.A.	HP	RPM	V / Ø	FLA	KW	V / Ø				TC (MBH)	SC (MBH)							
AHU-ST CU-ST	SEE PLAN	5.0	2000	300	1.5	1050	---	7.2	5.76	208/1	44	45	170	60	40	208/1	32	50	295	14.3	TRANE OR EQUAL	5TWR5060N1 TEM4B0C60S51
AHU-ST CU-ST	SEE PLAN	3.0	1200	200	1	735	---	5.3	5.0	208/1	40	40	166	36	22	208/1	18	30	235	14.3	TRANE OR EQUAL	5TTR5036 TEM4B0C37M31

NOTES:

- 1- UNITS RATED PER ARI 210, 240 AND 270.
- 2- PROVIDE WITH TXV, LIQUID LINE FILTER DRYER AND MULTI-USE SERVICE VALVES.
- 3- PROVIDE COMPRESSOR WITH CRANKCASE HEATER AND A MINIMUM 5-YEAR WARRANTY.
- 4- PROVIDE HIGH AND LOW PRESSURE CONTROL AND OVER TEMPERATURE PROTECTION.
- 5- PROVIDE WEATHERPROOF ELECTRIC CONTROLS AND SINGLE SIDE SERVICE ACCESS.
- 6- PROVIDE SINGLE POINT POWER ENTRY AND HEAVY DUTY NICKEL-CHROMIUM ELEMENT HEATER.
- 7- PROVIDE 1" THROWAWAY, MIN. 30% EFF. FILTER AND VIBRATION ISOLATION FOR AHU.
- 8- PROVIDE SPLIT FACE COIL WITH BOTTOM OF COIL ACTIVE FOR MULTIPLE CIRCUIT SYSTEM.
- 9- PROVIDE FACTORY MOUNTED FUSIBLE DISCONNECT/STARTER FOR AHU. COORDINATE PRIOR TO PURCHASING.
- 10- PROVIDE DISCONNECT FOR CU. (INSTALLED BY ELECTRICAL) COORDINATE PRIOR TO PURCHASING.
- 11- PROVIDE FACTORY PROGRAMMABLE THERMOSTAT TO MATCH CAPACITY STAGES.
- 12- PROVIDE REFRIGERANT LINE SIZE AS RECOMMENDED BY MANUFACTURER NOT TO EXCEED 120' EQUIV. LENGTH. FOR LONGER RUNS COORDINATE WITH MANUFACTURER PRIOR TO PURCHASE OR ANY WORK.
- 13- PROVIDE A LIQUID LINE SOLENOID VALVE INSTALLED AT THE AHU IF THE INSTALLED LINEAL LENGTH OF THE REFRIGERANT SUCTION LINE EXCEEDS 75 FT. (FIELD VERIFY)
- 14- PROVIDE A SUCTION LINE ACCUMULATOR IF THE INSTALLED LINEAL LENGTH OF THE REFRIGERANT SUCTION LINES EXCEED 100 FT. (FIELD VERIFY)
- 15- PROVIDE 3" AUXILIARY DRAIN PAN WITH AUTOMATIC SHUT OFF FLOAT SWITCH OR SENSOR SWITCH IN UNIT'S DRAIN PAN SECONDARY CONNECTION, EQUAL TO "WATER GARD". INSTALL CONDENSATE PUMP IF CONDENSATE DISCHARGE PIPE SLOPE CANNOT BE MAINTAINED.
- 17- PROVIDE LOW AMBIENT CONTROL & PROVIDE RAWAL DEVICE FOR MODULATING REFRIGERATION CAPACITY.
- 18- MUST COMPLY WITH 2023 FLORIDA ENERGY EFFICIENCY STANDARD
- 19- CONTRACTOR SHALL VERIFY BREAKERS/FUSES SIZES WITH ELECTRICAL AND MANUFACTURER CUT SHEET

TYPICAL SEQUENCE OF OPERATION

THE A/C UNIT SHALL BE STARTED AND STOPPED FROM A ROOM SENSOR WIRED THROUGH A WALL MOUNTED THERMOSTAT. ON CALL FOR COOLING, THE SENSOR SHALL ENERGIZE THE COMPRESSOR(S) TO MAINTAIN SET POINT. ON CALL FOR HEATING, THE THERMOSTAT SHALL ENERGIZE THE ELECTRIC HEAT IN STAGES TO MAINTAIN SET POINT. SUPPLY FAN SHALL RUN CONTINUOUSLY DURING OCCUPIED HOURS. OUTSIDE AIR MOTORIZED DAMPERS/ECONOMIZERS SHALL BE CLOSED DURING UNOCCUPIED HOURS.

CEILING MOUNTED EXHAUST FAN SCHEDULE

UNIT NO.	SERVING	TYPE	CFM	BLADE TYPE	SONES	TOTAL STATIC IN H2O	FAN RPM	MOTOR DATA				EHPA	SELECTION BASED ON:		REMARKS
								HP	RPM	VOLT	PH	MISSILE IMPACT	MANUFACTURER	MODEL	
EF-1	TOILET	INLINE DIRECTDRIVE	100	—	0.5	0.25	950	19.4 WATTS	950	110	1	NO	GREENHECK	SP-A110	1,2,3,4,5
EF-2	JANITOR	INLINE DIRECTDRIVE	70	—	0.5	0.25	950	19.4 WATTS	950	110	1	NO	GREENHECK	SP-A110	1,2,3,4,5

REMARKS: 1 PROVIDE BACKDRAFT DAMPER 2 PROVIDE DISCONNECT AT THE UNIT 3 PROVIDE SCR CONTROLLER ON ALL DIRECT DRIVE FANS 4 INTERNALLY LINE FAN FOR SOUND-PROOFING 5 PROVIDE VIBRATION ISOLATION HANGERS 6 PROVIDE SEPARATE WALL SWITCH
GENERAL COMMENTS: ALL INLINE FANS WITH EXTERIOR MOTORS (NOT IN THE AIRSTREAM) TO HAVE AN INSULATED & VENTILATED FACTORY MOUNTED FAN ENCLOSURE BOX

OUTDOOR AIR CALCULATIONS

ZONE IDENTIFICATION	OCCUPANCY CATEGORY	REQUIRED OUTDOOR AIR/PERSON (CFM/PER) Rp	NO. OF PEOPLE Pz "SEAT COUNT"	RpPz(CFM)	REQUIRED OUTDOOR AIR/UNIT AREA (CFM/SQ) Ra	ZONE FLOOR AREA (SF) A2 "NET AREA"	RaA2(CFM)	BREATHING ZONE OUTDOOR AIR FLOW RATE (CFM) Vbz	ZONE AIR VENTILATION EFFICIENCY Evz	ZONE MINIMUM AIRFLOW (CFM) Voz	SINGLE ZONE OUTDOOR AIR INTAKE FLOW (CFM) Vot = Voz	ACTUAL OUTDOOR AIR PROVIDED (CFM)
AHU-ST & AHU-ST	OFFICE BUSINESS	5	27	135	0.06	2500	150	285	0.8	360	360	500
										TOTAL	360	
										TOTAL PROVIDED		500

REMARKS: REQUIRED OUTDOOR VENTILATION AIR VALUES TAKEN FROM TABLE 403.3.1.1 PER SECTION 403 OF THE 2023 FLORIDA MECHANICAL CODE. VENTILATION FOR ACCEPTABLE INDOOR AIR QUALITY.

FLEXIBLE AIR DUCT NOTE

FLEXIBLE AIR CONNECTORS, BOTH METALLIC AND NONMETALLIC, SHALL BE TESTED IN ACCORDANCE WITH UL 181. SUCH CONNECTORS SHALL BE LISTED AND LABELED AS CLASS 0 OR CLASS 1 FLEXIBLE AIR CONNECTORS AND SHALL BE INSTALLED IN ACCORDANCE WITH 2023 FMC SECTION 304.1.

FLEXIBLE AIR CONNECTORS SHALL BE LIMITED IN LENGTH TO 14 FEET PER SECTION 603.6.2.1 CONNECTOR LENGTH.

FLEXIBLE AIR CONNECTORS SHALL NOT PASS THROUGH ANY WALL, FLOOR OR CEILING PER SECTION 603.6.2.2 CONNECTOR PENETRATION LIMITATIONS.

AIR BALANCE SCHEDULE

ZONE IDENTIFICATION	MAKE-UP AIR	EXHAUST AIR	AIR BALANCE
AHU-ST & AHU-ST	+500		+230
EF-1, EF-1 & EF-2		-270	

AIR DEVICE SCHEDULE

SYM	MFR. / CAT. NO.	TYPE	NECK	FRAME	FINISH	REMARKS
A	TITUS - OMNI	CD	0 - 125 CFM = 6" RND	12 x 12	BAKED WHITE ENAMEL	SEE REMARKS BELOW ALL NOTES
			130 - 150 CFM = 6" RND	24 x 24		
			160 - 275 CFM = 8" RND	24 x 24		
			280 - 390 CFM = 10" RND	24 x 24		
B	TITUS - 350FL	RG	SEE PLANS	24 x 24	BAKED WHITE ENAMEL	ALL NOTES
D	TITUS ML-38	LD	LINEAR 3-SLOT DIFFUSER	4'-0" LG.	PER ARCHITECT	ALL NOTES

DEVICE TYPE LEGEND:

CD = CEILING DIFFUSER

RG = RETURN GRILLE

NOTES/REMARKS:

- 1- PROVIDE CEILING DEVICES WITH PROPER FRAME TO MATCH CEILING OR WALL TYPE AS CALLED FOR BY ARCHITECT.
- 2- DEVICES SHALL BE PROVIDED WITH FACTORY FINISH.
- 3- AIR DEVICES SHALL BE 4-WAY THROW UNLESS NOTED OTHERWISE OR SHOWN ON PLANS WITH DIRECTIONAL ARROW(S). LINEAR DIFFUSERS DISCHARGE PATTERN AS INDICATED ON PLANS.
- 4- IF NECESSARY PROVIDE TOP HAT FOR GRILLES AND DIFFUSERS.
- 5- PROVIDE EQUALIZING GRID (TITUS MODEL EG). FOR DIFFUSERS TAPPED DIRECTLY FROM BOTTOM OF DUCT. PROVIDE OPPOSED BLADE DAMPER (TITUS MODEL D-57).
- 6- WHERE 3-WAY THROW IS NOTED, INSTALL BAFFLES IN THE NECK OF DIFFUSERS.
- 7- PROVIDE OPPOSED BLADE DAMPER FOR ALL REGISTERS.
- 8- ALL DIFFUSERS ARE TO BE PAINTED TO MATCH CEILING SURFACE. RE: COORDINATE W/ ARCHITECT FOR CEILING FINISH/MATERIAL SCHEDULES.
- 9- ALL OUTLET DEVICES LOCATED IN THE KITCHEN SHALL BE OF ALUMINUM CONSTRUCTION
- 10- TOPS OF ALL AIR OUTLET DEVICES SHALL BE COVERED WITH INSULATION
- 11- DIFFUSERS (TYPE I) WITHIN 10 FEET OF HOODS SHALL BE OF THE PERFORATED NON-DIRECTIONAL TYPE

MECHANICAL KEY PLAN NOTES: (X)

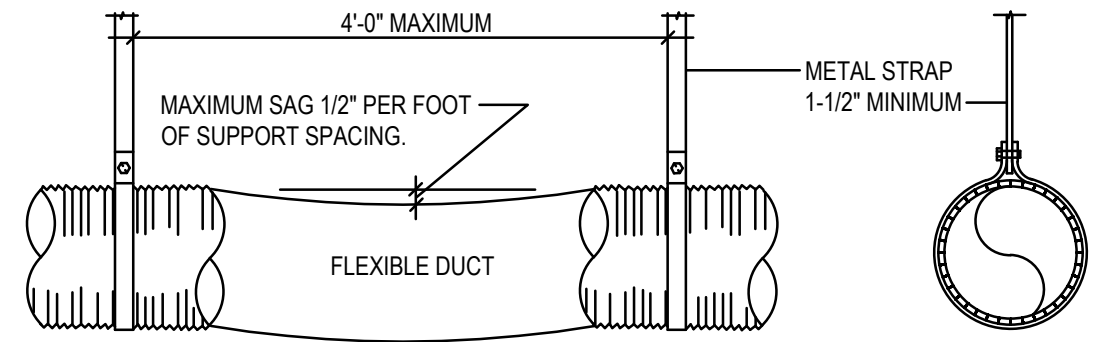
- 1- FLOOR MOUNTED VERTICAL AHU TO BE PROVIDED WITH A STRUCTURAL FRAME. INSTALL PER MANUFACTURER'S RECOMMENDATIONS. PROVIDE WITH VIBRATION ISOLATION AND INSTALL FLEXIBLE CONNECTIONS BETWEEN DUCTWORK AND FAN. REFRIGERANT PIPING TO RUN TOWARDS ACCU AT EXTERIOR. OUTSIDE AIR CONNECTION TO BE PROVIDED WITH MOTORIZED DAMPER/INTERLOCKED WITH CORRESPONDING AHU SET TO MINIMUM O.A. AS SCHEDULED. CONDENSATE DRAIN TO EXTERIOR AND SPILL TO LANDSCAPING/DRYWELL. SEE HVAC SHEET NOTES #4 & 5. AHU2 RETURN THRU LOUVERED CLOSET DOOR. COORDINATE DOOR WITH ARCHITECT.
- 2- DUCTWORK TO RUN HIGH AS POSSIBLE TIGHT TO THE UNDERSIDE OF STRUCTURE. OFFSET DUCT OR AHU LOCATION TO CLEAR THE ROOF LADDER. ALL DUCTS MUST BE INSULATED.
- 3- CONDENSATE DISCHARGE
- 4- RETURN AIR DUCT MOUNTED TEMPERATURE SENSORS.
- 5- THERMOSTAT TO BE MOUNTED 48" A.F.F. FINAL LOCATION TO BE COORDINATED WITH OWNER/ARCHITECT. THERMOSTATS FOR ALL AHU'S TO BE CENTRALLY LOCATED IN OFFICE AREA OR AS SHOWN ON DRAWINGS.
- 6- CEILING MOUNTED TOILET EXHAUST FAN. TOILET EXHAUST DUCT TO EXHAUST THROUGH EXTERIOR WALL TO BE TERMINATED WITH A WALL CAP OR EXHAUST LOUVER. TOILET EXHAUST DUCT UP TO ROOF TO BE TERMINATED WITH A GOOSENECK. CONTRACTOR TO COORDINATE INSTALLATION OF EXHAUST FAN WITH MANUFACTURER'S RECOMMENDATIONS. PROVIDE WITH VIBRATION ISOLATION, BACKDRAFT DAMPER AND INSTALL FLEXIBLE CONNECTIONS BETWEEN DUCTWORK AND FAN. SEE DETAIL ON SHEET M501.
- 7- 14"x12" OR 14"Ø OUTSIDE AIR DUCT UP TO ROOF GOOSENECK, COORDINATE EXACT SIZE AND LOCATION WITH ARCHITECT. FRESH AIR INTAKE MUST BE 10 FEET AWAY FROM ANY EXHAUST OUTLET. SEE DETAIL ON M502.
NOTE: 10"x10" OR 8"x8" OUTSIDE AIR DUCTS CONNECT TO RA MAIN. PROVIDE WITH A MOTORIZED DAMPER INTERLOCKED WITH CORRESPONDING AHU. VENT THRU EXTERIOR WALL LOUVER
- 8- MOUNT AIR COOLED CONDENSER C.U. ON 4" THICK CONCRETE PAD. PAD DIMENSIONS SHALL EXCEED CU DIMENSIONS BY 4" ON ALL SIDES AND SHALL ALLOW FOR REFRIGERANT PIPING STUB-UP. PLACE C.U. AWAY FROM ROOF DRIP LINE. PROVIDE MANUFACTURER'S RECOMMENDED CLEARANCES. ANCHOR C.U. SECURELY TO PAD. SEE DETAIL ON SHEET M502. LOCATION OF C.U. IS APPROXIMATE. COORDINATE CLOSELY WITH LANDSCAPING PLANS AND ARCHITECTURAL DETAILS. ROUTE REFRIGERANT LINES FROM AHU MECH CLOSET & CLOSET CHASE DOWN TO BELOW GRADE THROUGH CU PAD. INSULATE SUCTION LINE WITH MIN. 1/2" INSULATION. SEE UNDERGROUND REFRIGERANT LINE DETAIL ON SHEETS M50. WHEN REFRIGERANT LINE RUNS EXCEED 50 EQUIVALENT FEET, SIZE REFRIGERANT LINES PER CU MANUFACTURER'S RECOMMENDATIONS FOR THE SPECIFIC LENGTH INVOLVED. ALSO PROVIDE ACCESSORIES AS RECOMMENDED BY MANUFACTURER. IF THE SUBMITTED MANUFACTURER MAKES NO RECOMMENDATIONS FOR LONG REFRIGERANT LINE RUNS, CHOOSE ANOTHER CU MANUFACTURER.

MECHANICAL SHEET NOTES

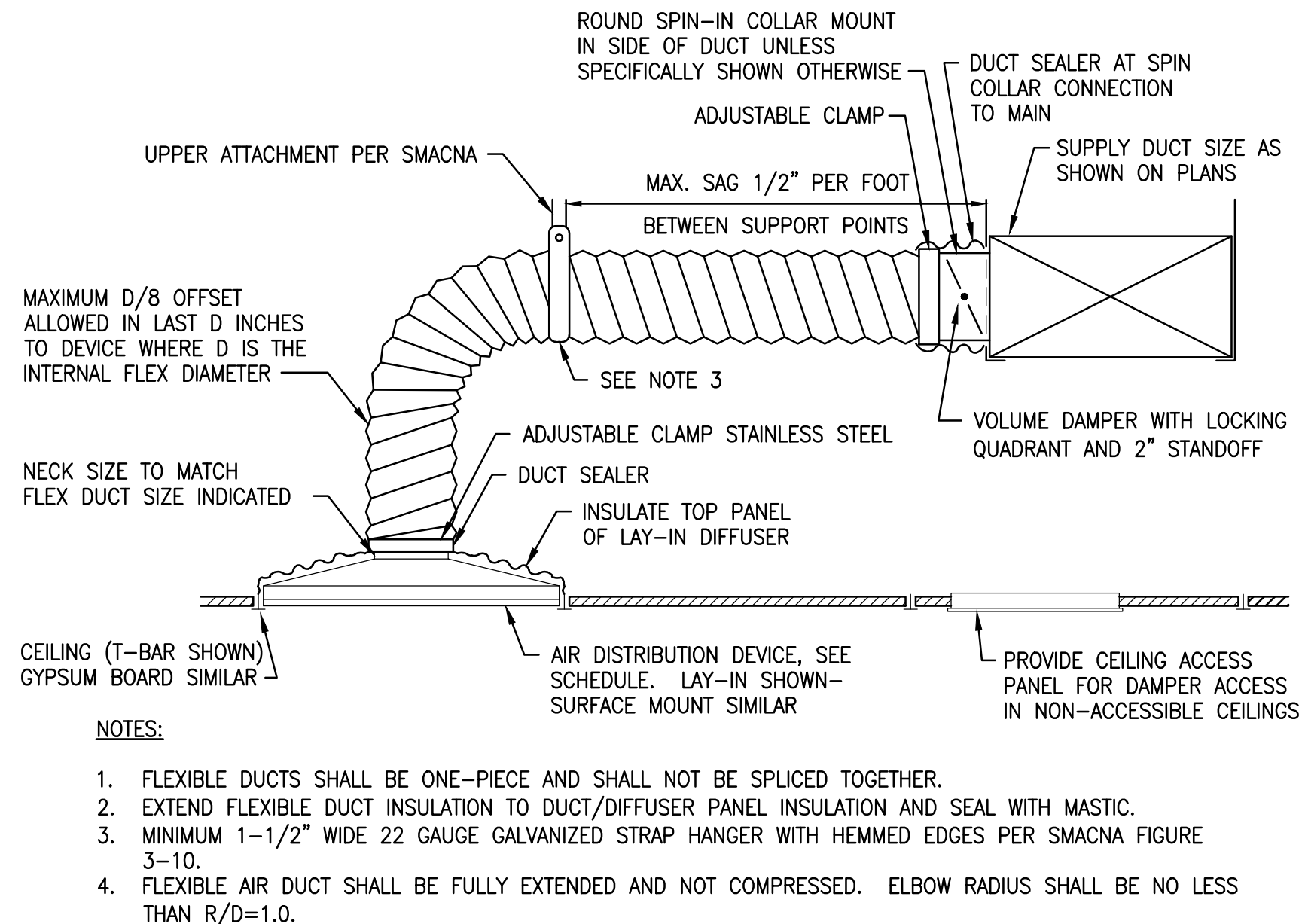
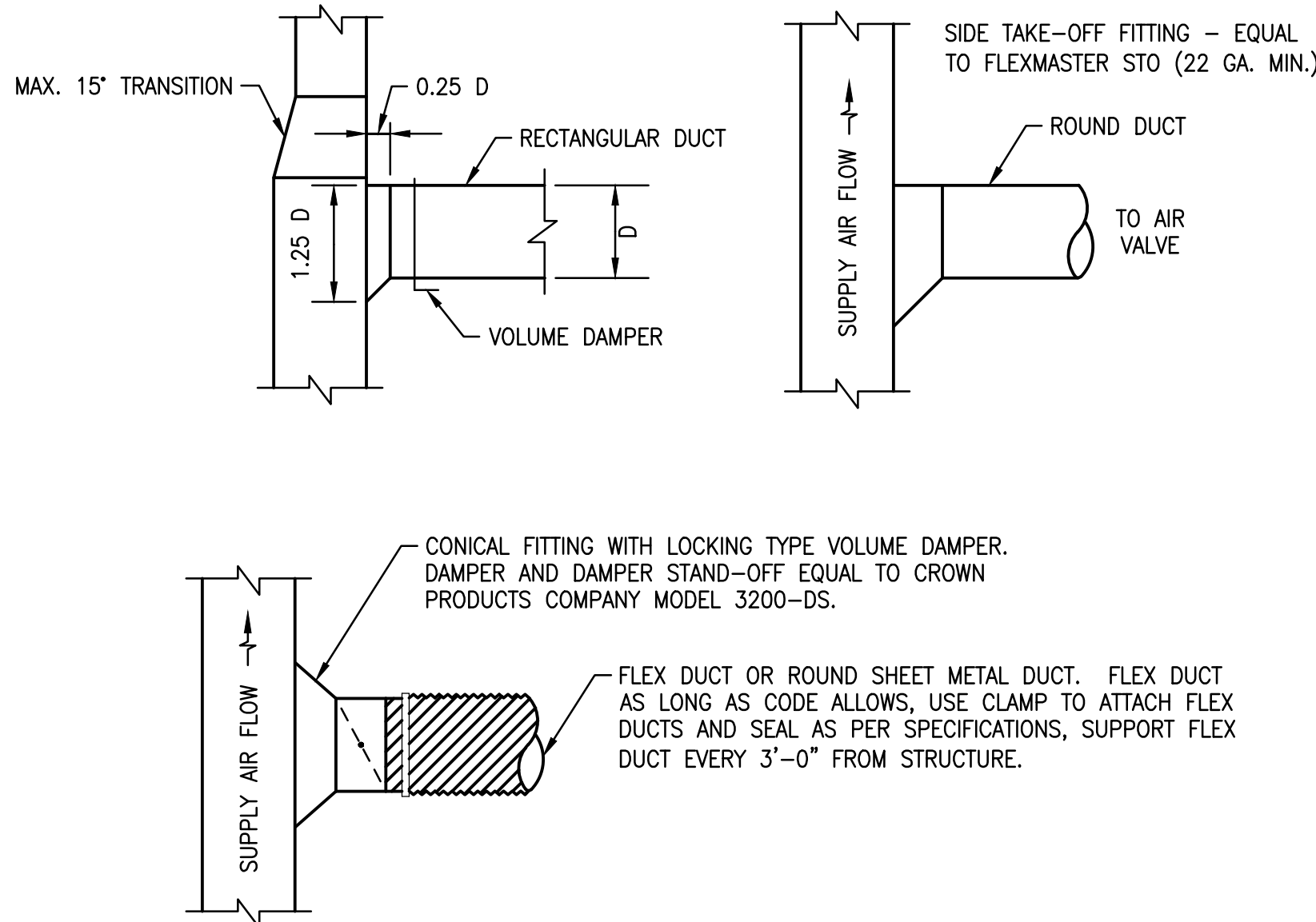
1. COMPLY WITH THE LATEST EDITIONS OF NFPA AND WITH 8TH EDITION OF THE 2023 FLORIDA MECHANICAL CODE.
2. SUFFICIENT SPACE SHALL BE PROVIDED ADJACENT TO ALL MECHANICAL COMPONENTS LOCATED IN OR FORMING A PART OF THE AIR DISTRIBUTION SYSTEM TO ASSURE ADEQUATE ACCESS FOR (1) CONSTRUCTION AND SEALING IN ACCORDANCE WITH THE REQUIREMENTS OF SECTION 603.1 OF THIS CODE; (2) INSPECTION; AND (3) CLEANING AND MAINTENANCE. A MINIMUM OF 4 INCHES IS CONSIDERED SUFFICIENT SPACE AROUND AIR HANDLING UNITS.
3. HVAC SHALL BE INSTALLED IN ACCORDANCE WITH NFPA 90A (STANDARD FOR THE INSTALLATION OF AIR CONDITIONING AND VENTILATING SYSTEM).
4. CONDENSATE FROM ALL COOLING COILS AND EVAPORATORS SHALL BE CONVEYED FROM THE DRAIN PAN OUTLET TO AN APPROVED PLACE OF DISPOSAL. SUCH PIPING SHALL MAINTAIN A MINIMUM HORIZONTAL SLOPE IN THE DIRECTION OF DISCHARGE OF NOT LESS THAN ONE-EIGHTH UNIT VERTICAL IN 12 UNITS HORIZONTAL (1-PERCENT SLOPE). CONDENSATE SHALL NOT DISCHARGE INTO A STREET, ALLEY OR OTHER AREAS SO AS TO CAUSE A NUISANCE. DRAIN INTO DRYWELL
5. A WATER-LEVEL DETECTION DEVICE CONFORMING TO UL 508 SHALL BE PROVIDED THAT WILL SHUT OFF THE EQUIPMENT SERVED IN THE EVENT THAT THE PRIMARY DRAIN IS BLOCKED. THE DEVICE SHALL BE INSTALLED IN THE PRIMARY DRAIN LINE, THE OVERFLOW DRAIN LINE, OR IN THE EQUIPMENT-SUPPLIED DRAIN PAN, LOCATED AT A POINT HIGHER THAN THE PRIMARY DRAIN LINE CONNECTION AND BELOW THE OVERFLOW RIM OF SUCH PAN. THIS DEVICE SHALL SHUT OFF THE EQUIPMENT SERVED IN THE EVENT THAT THE PRIMARY DRAIN BECOMES RESTRICTED. DEVICES INSTALLED IN THE DRAIN LINE SHALL NOT BE PERMITTED.
6. REFRIGERANT LINES. ROUTE FROM AHU MECH CLOSET CHASE DOWN TO BELOW GRADE THROUGH CU PAD OR TO ROOF MOUNTED CU. INSULATE SUCTION LINE WITH MIN. 1/2" INSULATION. SEE UNDERGROUND REFRIGERANT LINE DETAIL ON SHEETS M501/502. CONDENSATE NOTE, DISCHARGE AND TERMINATE EACH AHU'S CONDENSATE WATER INTO DRYWELL
7. WHEN REFRIGERANT LINE RUNS EXCEED 50 EQUIVALENT FEET, SIZE REFRIGERANT LINES PER CU MANUFACTURER'S RECOMMENDATIONS FOR THE SPECIFIC LENGTH INVOLVED. ALSO PROVIDE ACCESSORIES AS RECOMMENDED BY MANUFACTURER. IF THE SUBMITTED MANUFACTURER MAKES NO RECOMMENDATIONS FOR LONG REFRIGERANT LINE RUNS, CHOOSE ANOTHER CU MANUFACTURER.
8. ROUTE MINIMUM 26 GAUGE 4" ROUND DUCT FROM CEILING MOUNT EXHAUST FAN, THROUGH FLOOR/CEILING ASSEMBLY TO EXTERIOR PLASTIC WALL CAP WITH WEATHER HOOD AND BIRD SCREEN. PRIME COAT AND PAINT WALL CAP TO MATCH BUILDING. SEE DETAIL ON SHEETS M501/502. INSTALL EXHAUST FAN WITH BACKDRAFT DAMPER.
9. DUCT REGISTERS, GRILLES AND DIFFUSERS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S INSTALLATION INSTRUCTIONS. VOLUME DAMPERS OR OTHER MEANS OF SUPPLY AIR ADJUSTMENT SHALL BE PROVIDED AT EACH INDIVIDUAL DUCT REGISTER, GRILLE OR DIFFUSER. EACH VOLUME DAMPER OR OTHER MEANS OF SUPPLY AIR ADJUSTMENT USED IN BALANCING SHALL BE PROVIDED WITH ACCESS.
10. CONTRACTOR SHALL MAINTAIN A MINIMUM DISTANCE OF 10'-0" FROM EXHAUST AIR OUTLETS TO OUTSIDE AIR INTAKES.

1 HVAC Plan
1/4"=1'-0"

Ref. Dwg: M101



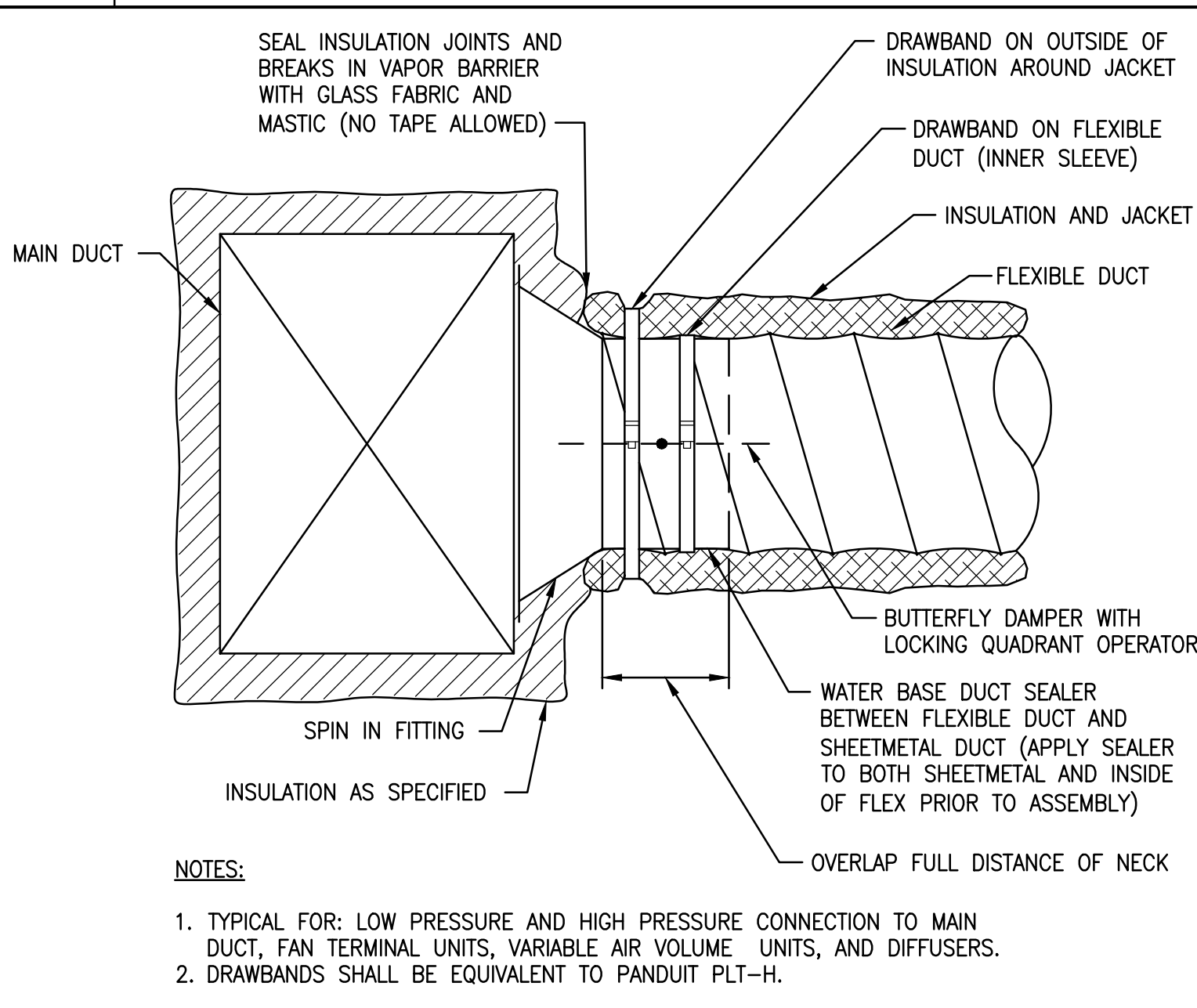
- NOTES:
- DUCT SHOULD EXTEND STRAIGHT FOR SEVERAL INCHES FROM A CONNECTION BEFORE BENDING.
 - SUPPORT SYSTEM MUST NOT DAMAGE DUCT OR CAUSE OUT OF ROUND SHAPE.



- NOTES:
- FLEXIBLE DUCTS SHALL BE ONE-PIECE AND SHALL NOT BE SPLICED TOGETHER.
 - EXTEND FLEXIBLE DUCT INSULATION TO DUCT/DIFFUSER PANEL INSULATION AND SEAL WITH MASTIC.
 - MINIMUM 1-1/2" WIDE 22 GAUGE GALVANIZED STRAP HANGER WITH HEMMED EDGES PER SMACNA FIGURE 3-10.
 - FLEXIBLE AIR DUCT SHALL BE FULLY EXTENDED AND NOT COMPRESSED. ELBOW RADIUS SHALL BE NO LESS THAN R/D=1.0.

1 FLEX DUCT SUPPORT

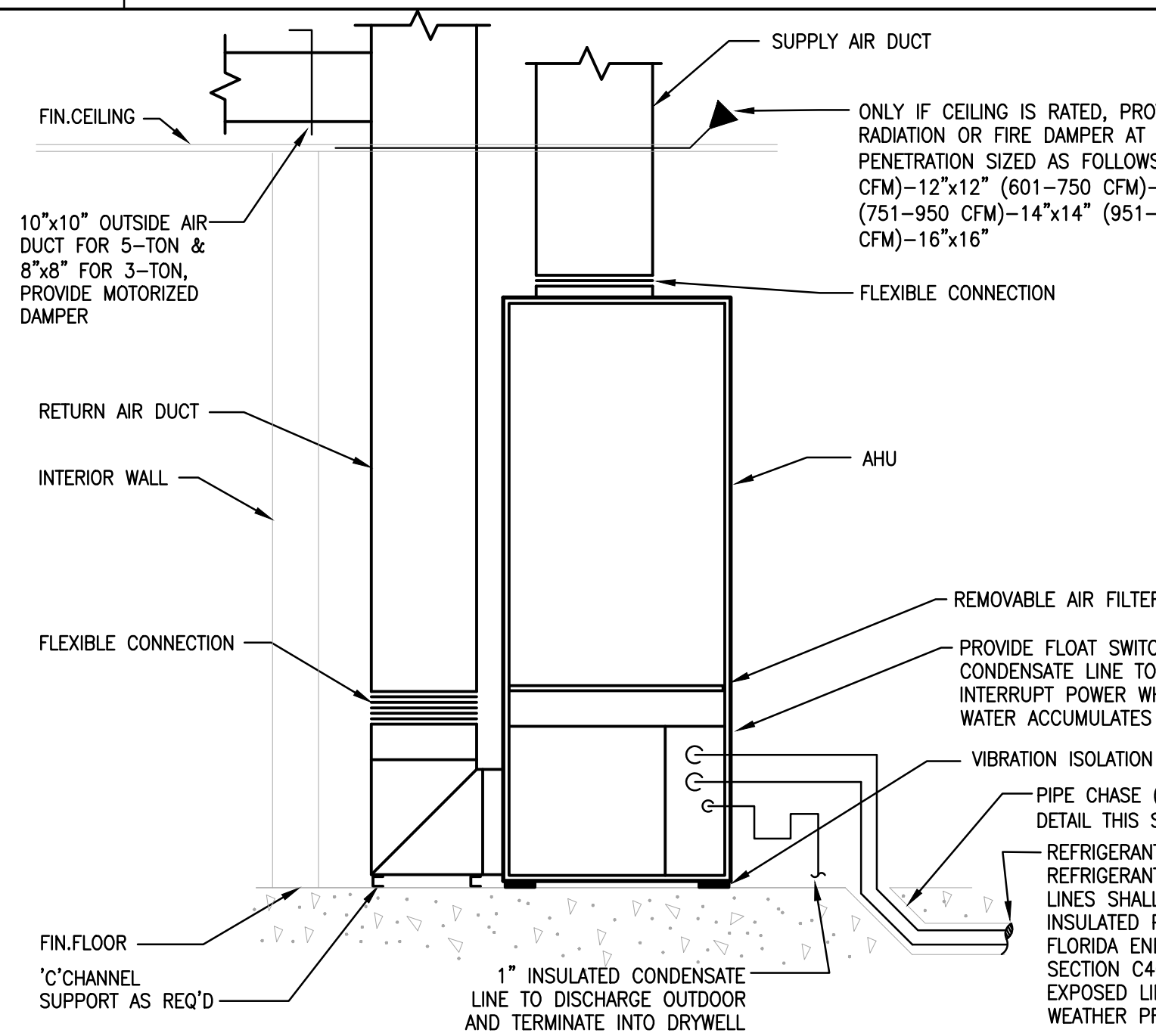
NTS



- NOTES:
- TYPICAL FOR: LOW PRESSURE AND HIGH PRESSURE CONNECTION TO MAIN DUCT, FAN TERMINAL UNITS, VARIABLE AIR VOLUME UNITS, AND DIFFUSERS.
 - DRAWBANDS SHALL BE EQUIVALENT TO PANDUIT PLT-H.

4 FLEXIBLE DUCT CONNECTION DETAIL

NTS

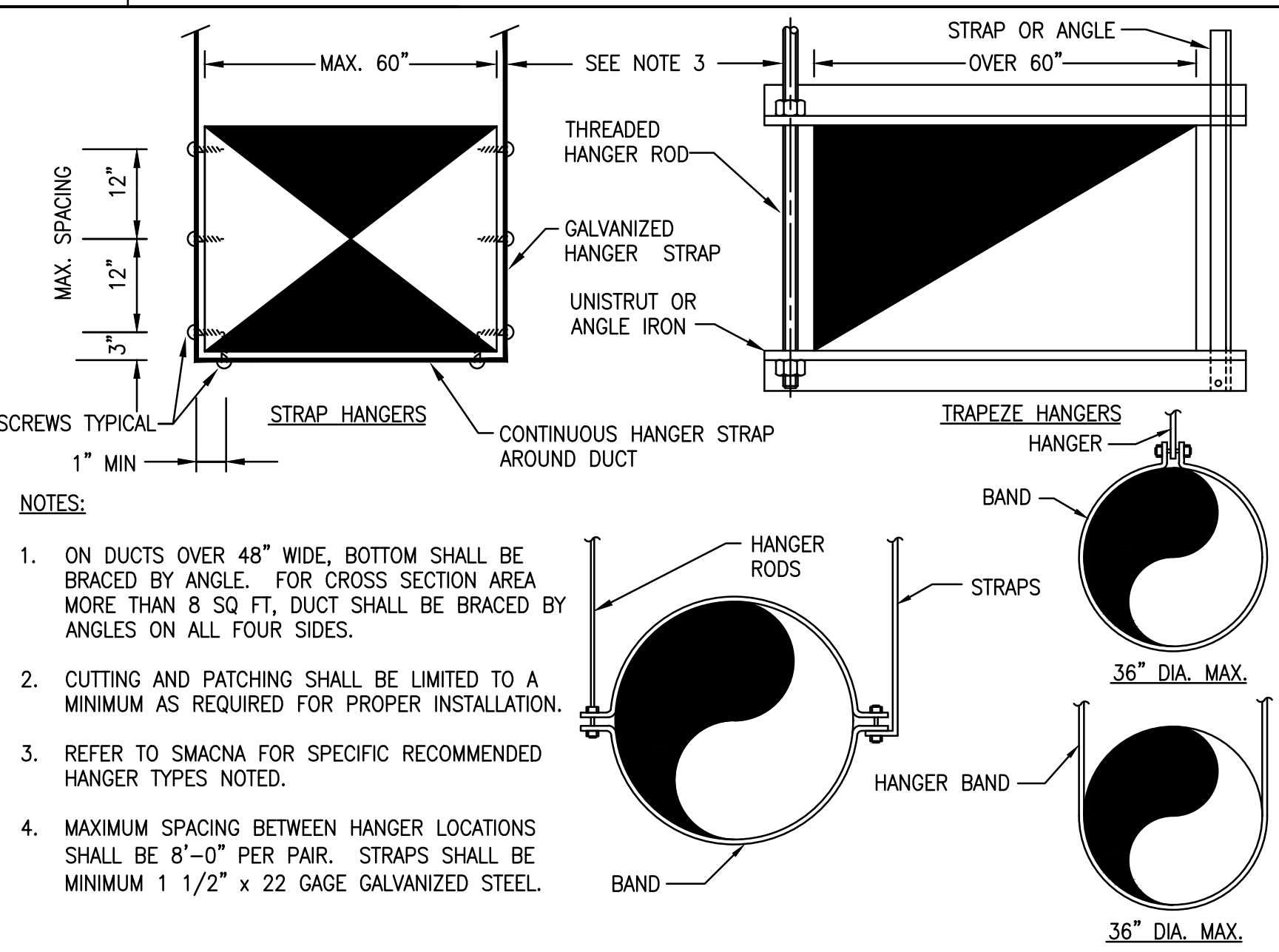


8 VERTICAL AHU MOUNTING DETAIL

NTS

2 DUCT CONNECTION DETAILS

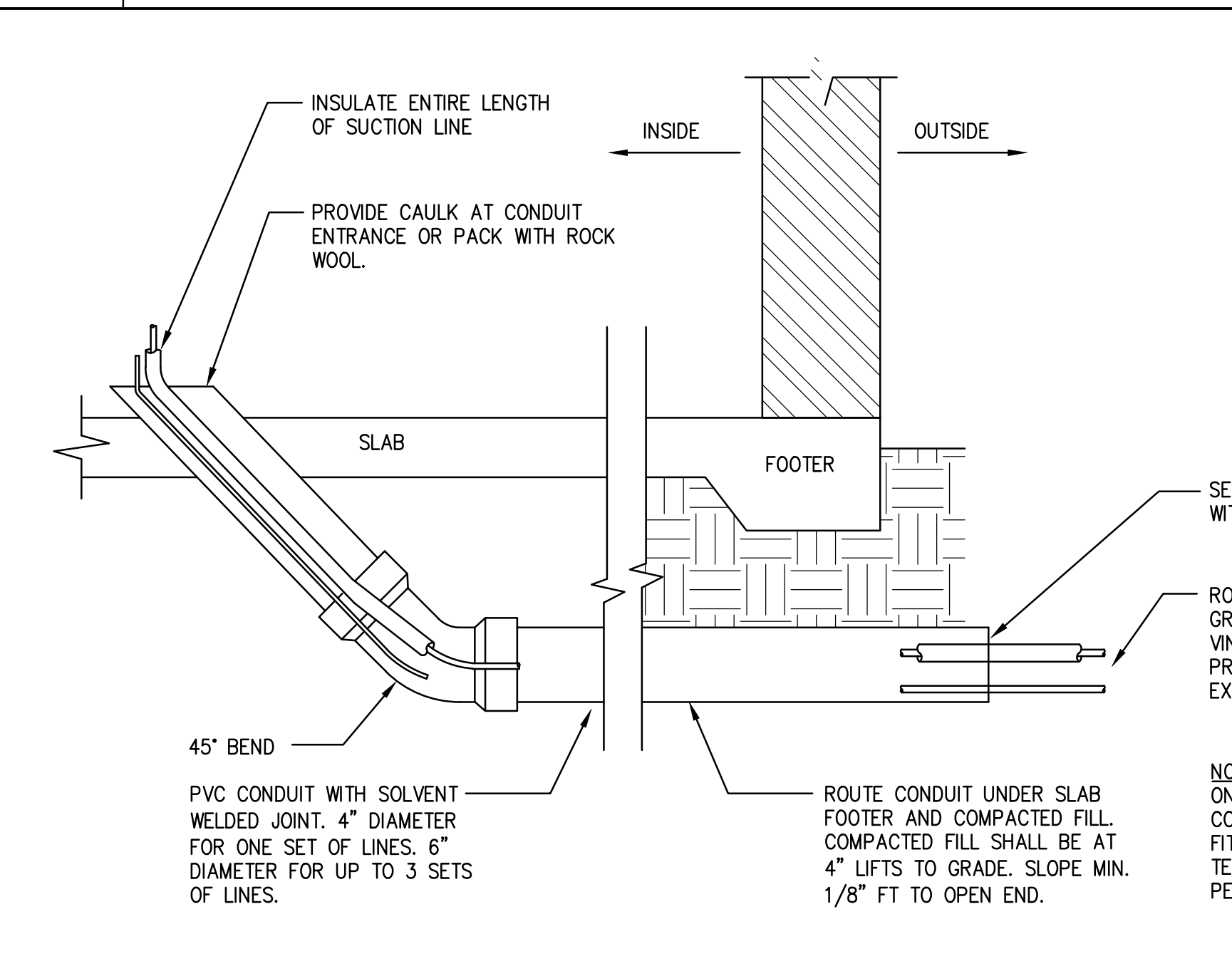
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- NOTES:
- ON DUCTS OVER 48" WIDE, BOTTOM SHALL BE BRACED BY ANGLE. FOR CROSS SECTION AREA MORE THAN 8 SQ FT, DUCT SHALL BE BRACED BY ANGLES ON ALL FOUR SIDES.
 - CUTTING AND PATCHING SHALL BE LIMITED TO A MINIMUM AS REQUIRED FOR PROPER INSTALLATION.
 - REFER TO SMACNA FOR SPECIFIC RECOMMENDED HANGER TYPES NOTED.
 - MAXIMUM SPACING BETWEEN HANGER LOCATIONS SHALL BE 8'-0" PER PAIR. STRAPS SHALL BE MINIMUM 1 1/2" x 22 GAGE GALVANIZED STEEL.

5 TYPICAL HANGER DETAIL

NTS

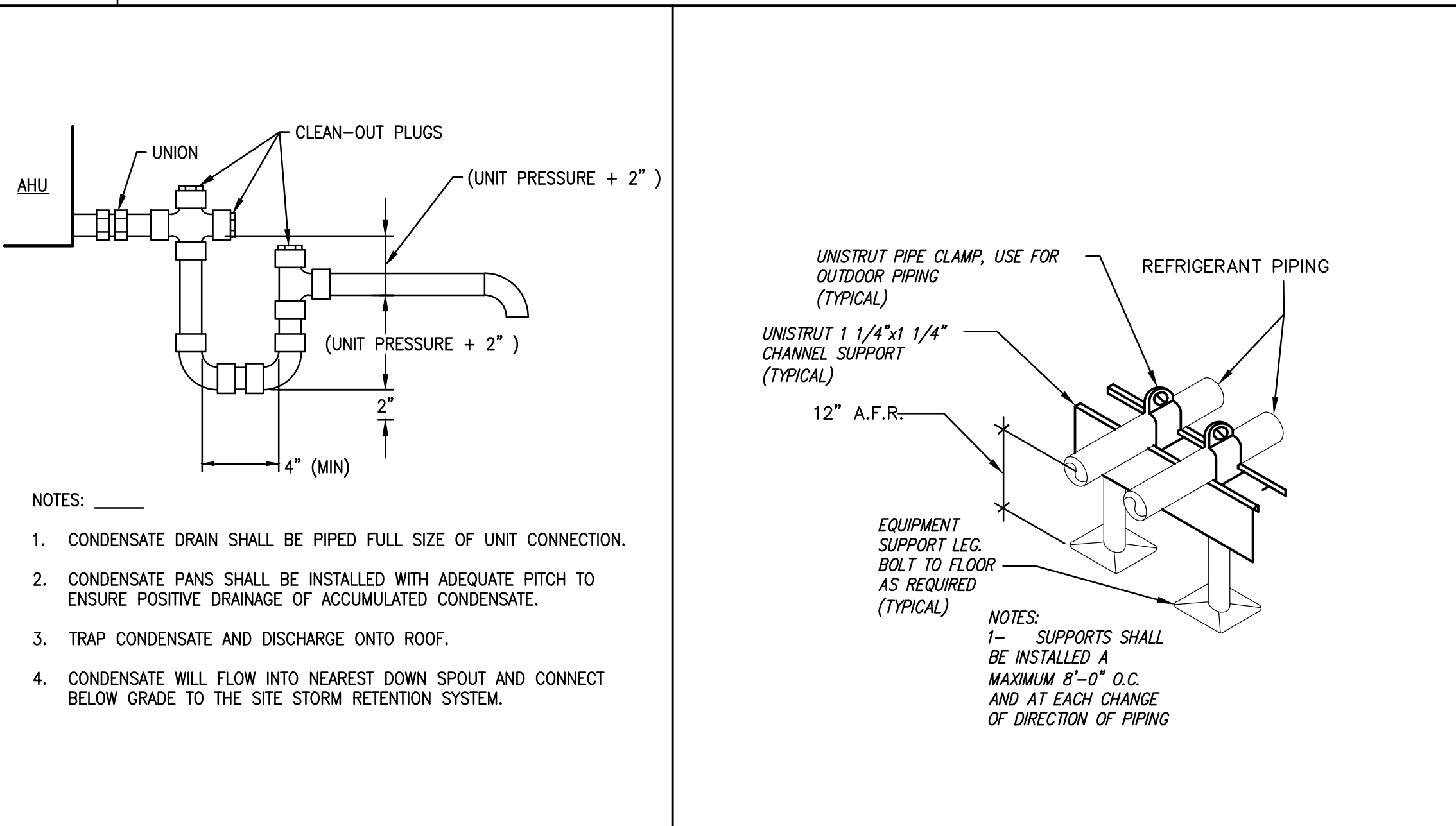


9 UNDERGROUND REFRIGERANT LINES

NTS

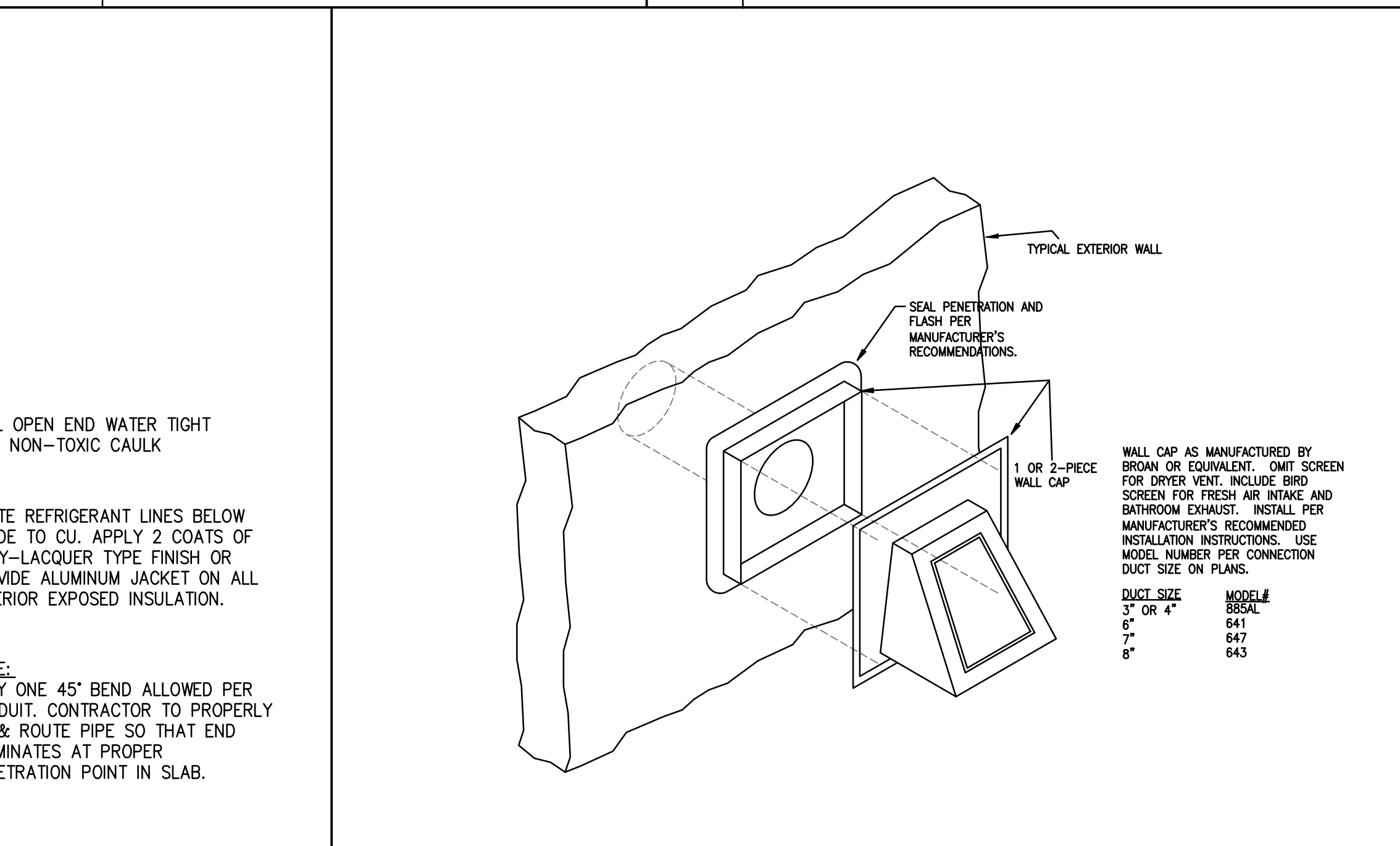
3 FLEXIBLE DUCT INSTALLATION DETAIL

NTS



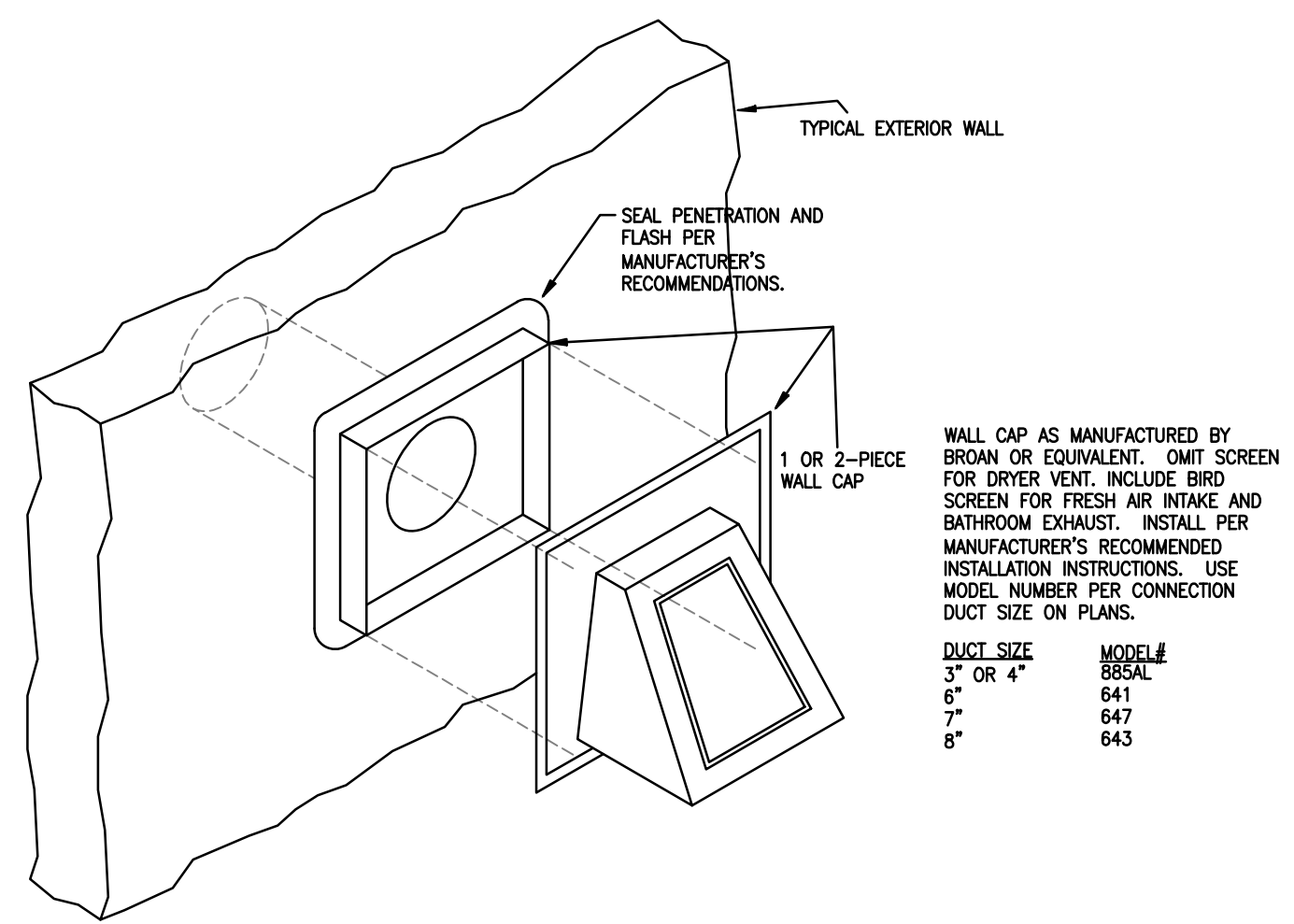
6 CONDENSATE DRAIN DETAIL

NTS

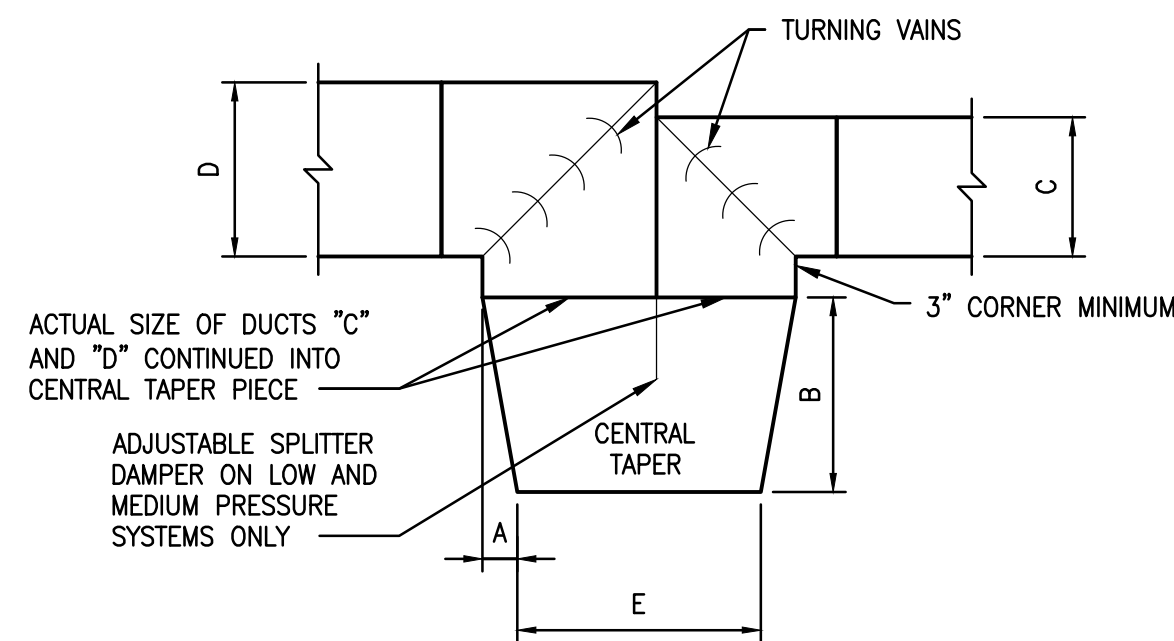


7 REFRIGERANT PIPING SUPPORT DETAIL

NTS

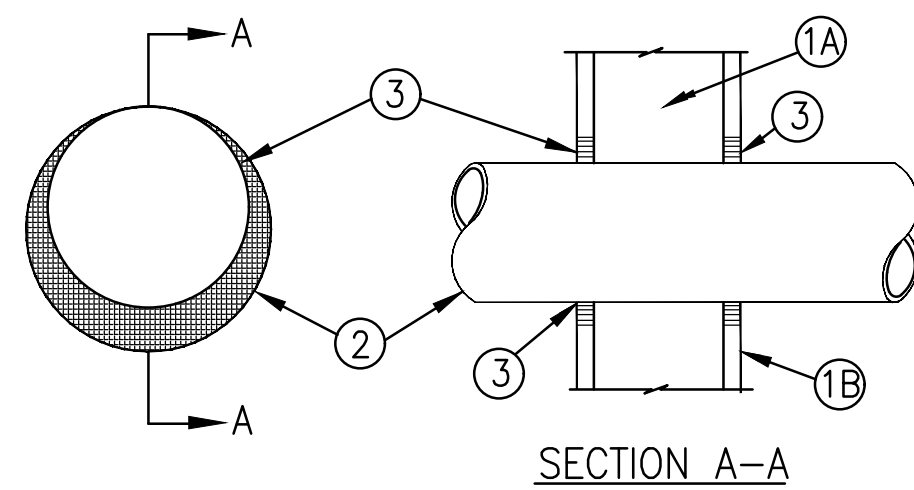


DUCT SIZE	MODEL#
3" or 4"	885AL
6"	641
7"	647
8"	643



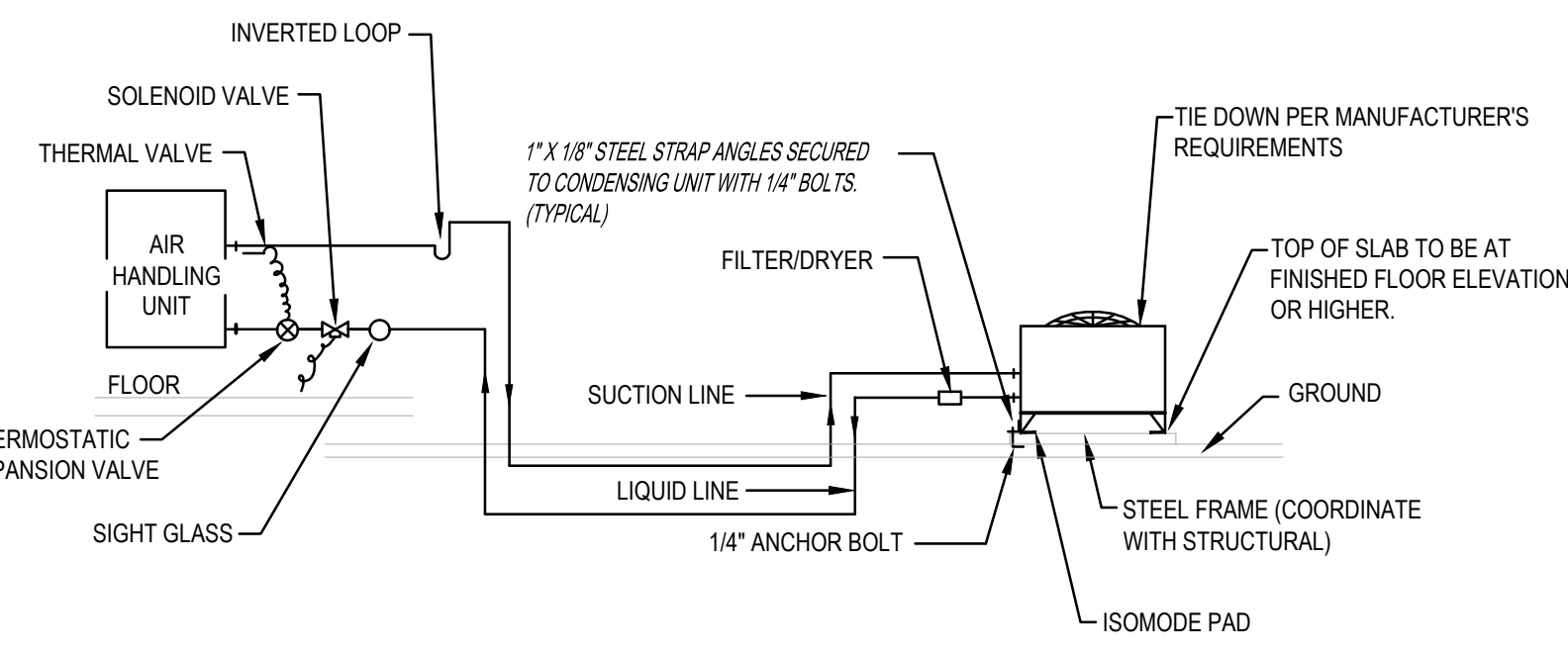
NOTES:

1. DIMENSION "A" IN INCHES, DIMENSION "B" IN FEET.
2. DEPTH OF "C" AND "D" WILL BE THE SAME.
3. PROPORTION SPLIT ACCORDING TO CFM IN EACH BRANCH.
4. OMIT SPLITTER DAMPER IN DUCTWORK BETWEEN AHU SUPPLY FAN AND TERMINAL BOX INLET.

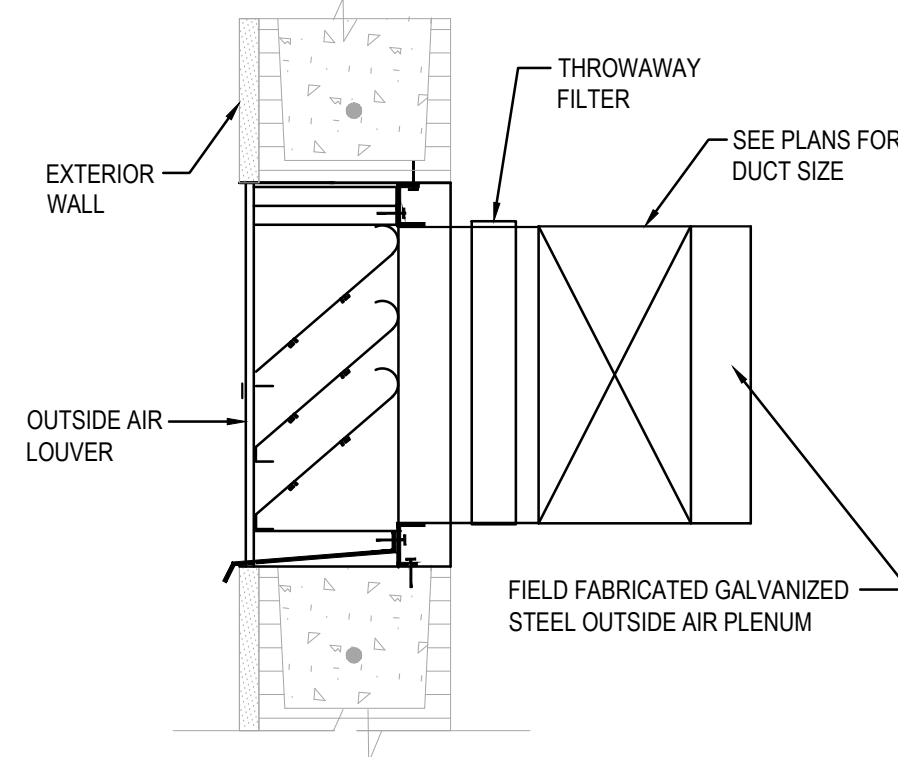


System No. W-L-7032
F Ratings--1 Hr (See Items 2 and 3)
T Ratings--0 Hr (See item 2 and 3)
L Rating at Ambient--less than 1 CFM/sq ft
L Rating At 400 F--less than 1 CFM/sq ft

1. WALL ASSEMBLY—THE 1 HR FIRE-RATED GYPSUM WALLBOARD/STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL AND PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
 - A. STUDS—WALL FRAMING SHALL CONSIST OF EITHER WOOD STUDS OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOM 2 BY 4 IN LUMBER SPACED 16 IN. OC. STEEL STUDS TO MIN 5/8 IN. THICK, 4 FT. WIDE BY 1-3/8 IN. DEEP CHANNELS SPACED MAX 24 IN. OC.
 - B. WALLBOARD, GYPSUM—ONE LAYER OF MIN. 5/8 IN. 3/12, 4 FT. WIDE WITH SQUARE OR TAPERED EDGES, THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAX DIA OF OPENING IS 7 IN.
2. STEEL VENT DUCT—NOM 8 IN. DIAM (OR SMALLER) SCHEDULE 18 (OR HEAVIER) GALVANIZED STEEL DUCT TO BE INSTALLED EITHER CONCENTRICALLY OR ECCENTRICALLY WITH THE FIRE STOP SYSTEM, AN ANNUAL SPACE OF MIN. 0 IN. (POINT CONTACT) TO MAX. 1 IN. IS REQUIRED WITH IN THE FIRE STOP SYSTEM. DUCT TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF THE WALL ASSEMBLY.
3. FILL, VOID OR CAVITY MATERIAL—CAULK OR SEALANT—MIN 8 IN. THICKNESS OF FILL MATERIAL, APPLIED WITH ANNULUS, FLUSH WITH BOTH SURFACES OF WALL ASSEMBLY AT THE POINT OF CONTACT LOCATION BETWEEN DUCT AND WALLBOARD. MIN 1/2 IN. DIAM BEAD OF SEALANT SHALL BE APPLIED AT THE WALL BOARD/DUCT INTERFACE ON BOTH SURFACES OF THE WALL ASSEMBLY.



NOTE:
PIPING SIZED AS PER MANUFACTURER'S RECOMMENDATIONS WITH CONSIDERATION FOR DISTANCE OF PIPING AND HEIGHT OF RISERS.



1	SPLITTER DETAIL	2	THRU WALL DUCT PENETRATION DETAIL										
NTS		NTS											
3	CONDENSATE DRYWELL DETAIL	4	CEILING MOUNTED EXHAUST FAN DETAIL										
NTS	<table><tr><th colspan="2">FASTENER QUANTITY TABLE</th></tr><tr><th>LENGTH OF SIDE</th><th>QUANTITY</th></tr><tr><td>LESS THAN 12"</td><td>1 PER SIDE</td></tr><tr><td>12" TO 24"</td><td>2 PER SIDE</td></tr><tr><td>24" TO 36"</td><td>3 PER SIDE</td></tr></table>	FASTENER QUANTITY TABLE		LENGTH OF SIDE	QUANTITY	LESS THAN 12"	1 PER SIDE	12" TO 24"	2 PER SIDE	24" TO 36"	3 PER SIDE	NTS	
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LENGTH OF SIDE	QUANTITY												
LESS THAN 12"	1 PER SIDE												
12" TO 24"	2 PER SIDE												
24" TO 36"	3 PER SIDE												
5	GROUND MOUNTED EQUIPMENT TIE DOWN	9	OUTSIDE AIR LOUVER GOOSENECK DETAIL										
NTS		NTS											

ELECTRICAL SPECIFICATIONS

GENERAL

1. ALL WORK SHALL BE IN ACCORDANCE WITH THE LATEST ADOPTED EDITION OF THE FLORIDA BUILDING CODE (FBC), NFPA 101, ALL LOCAL CODES, ORDINANCES, REGULATIONS AND UTILITY POWER AND TELEPHONE COMPANY STANDARDS.

2. ALL BUILDING WIRE SHALL BE COPPER TYPE "THHN/THWN" FOR ALL SIZES. MINIMUM WIRE SIZE SHALL BE #12 AWG.

3. ALL BRANCH AND FEEDER CIRCUITS SHALL CONTAIN A SEPARATE GROUNDING CONDUCTOR AND SEPARATE NEUTRAL CONDUCTOR AND SHALL BE SIZED AND BONDED IN ACCORDANCE WITH ARTICLE 250 OF THE N.E.C.

4. ALL CONDUIT INSTALLED IN INTERIOR LOCATIONS SHALL BE TYPE E.M.T. WITH SET SCREW STEEL CONNECTORS AND COUPLINGS. ALL EMT SHALL BE A MINIMUM 1/2". ALL CONDUIT INSTALLED IN EXTERIOR LOCATIONS, ABOVE GRADE, SHALL BE GALVANIZED RIGID CONDUIT. ALL CONDUIT BELOW GRADE SHALL BE SCHEDULE 40 P.V.C. ALL CONDUIT SHALL BE INSTALLED CONCEALED. THE USE OF METAL CLAD CABLE (MC) SHALL BE PERMITTED ON THIS PROJECT.

5. CONTRACTOR SHALL COORDINATE THE LOCATION OF RECEPTACLES, CONTROL CIRCUITS, COMMUNICATIONS AND DATA OUTLETS, LIGHTING FIXTURES AND DEVICES WITH THE OWNER'S REPRESENTATIVE PRIOR TO INSTALLATION.

6. DRAWINGS ARE DIAGRAMMATIC – CONTRACTOR SHALL REFER TO ARCHITECTURAL DRAWINGS AND FIELD CONDITIONS FOR ALL DIMENSIONS.

7. CONTRACTOR SHALL OBTAIN AND FURNISH ALL PERMITS REQUIRED. PROVIDE ALL LABOR, MATERIALS, EQUIPMENT, APPLIANCES AND TOOLS TO PERFORM ALL WORK NECESSARY FOR THE COMPLETE EXECUTION OF THE ELECTRICAL WORK AS SHOWN ON THE DRAWINGS. PROVIDE WORK NOT SPECIFICALLY SHOWN OR SPECIFIED, YET REQUIRED TO INSURE PROPER AND COMPLETE OPERATION OF ALL SYSTEMS AND TO SATISFY THE DESIGN INTENT OF THE WORK AND COMPLY WITH ALL APPLICABLE CODES AND REGULATIONS.

8. THE DRAWINGS INDICATE THE GENERAL ARRANGEMENT OF CIRCUITS, OUTLETS, LOCATIONS OF SWITCHES, PANELBOARDS, CONDUITS AND OTHER WORK. PRIOR FIELD VERIFICATION OF ALL DIMENSIONS IS REQUIRED AS PART OF THIS WORK. CONDUIT RUNS AND GROUNDING ARE SHOWN DIAGRAMMATICALLY ONLY. FIELD VERIFY THE ACTUAL ROUTING OF CONDUITS PRIOR TO BIDDING.

9. PROVIDE ALL 125VAC 20 AMP SPECIFICATION GRADE RATED WIRING DEVICES UNLESS OTHERWISE NOTED. PROVIDE MATCHING COVER PLATES, JUNCTION BOXES AND PLASTER RINGS AS REQUIRED TO FACILITATE THE INSTALLATION OF FLUSH MOUNTED WIRING DEVICES AS NOTED AND AS THE PROJECT CONDITIONS WARRANT.
10. ANY PENETRATIONS MADE THROUGH A FIRE RATED ASSEMBLY SHALL BE PROPERLY SEALED TO MAINTAIN THE INTEGRITY OF THE ASSEMBLY PER U.L. AND NFPA.

11. CONTRACTOR SHALL PROVIDE STRICT COORDINATION WITH NEW EQUIPMENT AND FIELD VERIFY THE EXACT FINAL LOCATION AND EQUIPMENT POWER CHARACTERISTICS WITH THE OWNER'S AUTHORIZED REPRESENTATIVE AND OTHER TRADES PRIOR TO CONSTRUCTION.

12. EACH BIDDER SHALL INSPECT THE SITE AS REQUIRED FOR KNOWLEDGE OF EXISTING CONDITIONS PRIOR TO BIDDING AND FAILURE TO OBTAIN SUCH KNOWLEDGE SHALL NOT RELIEVE THE SUCCESSFUL BIDDER OF THE RESPONSIBILITY TO COORDINATE EXISTING CONDITIONS IN PERFORMING THE WORK UNDER THIS CONTRACT.

13. WHERE NEW WORK CANNOT BE INSTALLED WITHOUT CHANGES IN EXISTING FACILITIES OR SYSTEMS OR WHERE IT IS INDICATED ON DRAWINGS TO REWORK AN EXISTING INSTALLATION, THIS CONTRACT SHALL INCLUDE ALTERATIONS TO EXISTING WORK AS REQUIRED TO INSTALL NEW WORK. ADDITIONS TO THE CONTRACT COST WILL NOT BE ALLOWED DUE TO THE CONTRACTORS FAILURE TO INSPECT EXISTING CONDITIONS AT THE SITE OF THE WORK.

14. PROVIDE HACR RATED CIRCUIT BREAKERS FOR ALL HVAC EQUIPMENT.

15. THE WORK SHALL INCLUDE REVISIONS, MODIFICATIONS, AND REWORK OF THE EXISTING FACILITY ELECTRICAL SYSTEMS AS REQUIRED FOR INSTALLATION OF NEW WORK, AND FOR CONNECTIONS BETWEEN EXISTING WORK AND NEW WORK WHERE REQUIRED. THE WORK SHALL ALSO INCLUDE THE COMPLETION OF ELECTRICAL POWER AND CONTROL CIRCUITS FOR DEVICES AND EQUIPMENT THAT ARE TO REMAIN IN SERVICE. IF THE CIRCUITS ARE BROKEN BY DEMOLITION WORK, OR BY THE REMOVAL OR CUTTING OF EXISTING BUILDING CONSTRUCTION, THE EXISTING CONDUIT AND WIRING SHALL BE REROUTED AND RECONNECTED WHERE NECESSARY.

16. PROVIDE PANELBOARD MATCHING CIRCUIT BREAKERS IN ORDER TO CONNECT NEW BRANCH CIRCUITS. UPDATE ALL PANELBOARD DIRECTORIES TO REFLECT THE ACTUAL CONDITIONS. ALL UPDATES SHALL BE TYPEWRITTEN. ALL NEW ELECTRICAL DEVICES MUST BE LABELED AT THE DEVICE WITH CIRCUIT AND PANELBOARD NUMBER.

CUTTING AND PATCHING

1. THE RESPONSIBILITY FOR ANY CUTTING OF CONSTRUCTION WHICH IS REQUIRED FOR THE INSTALLATION OF DIVISION 16 WORK, SHALL BE BY THE CONTRACTOR. THE CONTRACTOR SHALL COORDINATE WITH ALL OTHER TRADES AND THE OWNER BEFORE ANY CUTTING AND OBTAIN APPROVAL FROM THE ARCHITECT/ENGINEER PRIOR TO ANY CUTTING. ALL PATCHING, PAINTING AND FINISH SHALL BE BY THE CONTRACTOR.

AS-BUILT DRAWINGS

2. CUTTING SHALL BE DONE WITH EXTREME CARE AND IN SUCH A MANNER THAT THE STRENGTH OF THE STRUCTURE WILL NOT BE ENDANGERED. WHEREVER POSSIBLE, OPENINGS IN CONCRETE OR MASONRY CONSTRUCTION SHALL BE BY CONCRETE SAW OR ROTARY CORE DRILL. OPENINGS IN ANY CONSTRUCTION SHALL BE CUT THE MINIMUM SIZE REQUIRED FOR THE INSTALLATION OF THE WORK. ADEQUATE PROTECTION SHALL BE PROVIDED TO PREVENT DAMAGE TO ADJACENT AREAS AND TO PREVENT DUST FROM SPREADING TO ADJACENT AREAS.

3. WHERE OPENINGS OR HOLES ARE CUT IN CONSTRUCTION AND THE CUTTING BREAKS ELECTRICAL CIRCUITRY OR CONTROL CIRCUITRY CONDUIT AND WIRING, THEN IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO REROUTE THE CIRCUITRY CONDUIT AND REWIRING AND TO COMPLETE THE CIRCUITRY AS REQUIRED AND AS APPROVED BY THE ARCHITECT/ENGINEER. TEMPORARY COMPLETION SHALL BE PROVIDED WHERE NECESSARY BEFORE THE PERMANENT REROUTING AND COMPLETION WORK IS FINISHED.

4. BEFORE ANY CUTTING, PATCHING, OR FINISHING WORK IS STARTED, DUST AND MOISTURE PROTECTION SHALL FIRST BE INSTALLED AS REQUIRED AND AS SPECIFIED IN THESE SPECIFICATIONS.

5. OPENINGS CUT IN FLOOR SHALL BE CUT BY CORE DRILLING WHERE POSSIBLE. AFTER WORK IS INSTALLED THROUGH ANY OPENING IN FLOOR, THE OPENING AROUND THE WORK SHALL BE PATCHED AND SEALED WATERTIGHT AND EPOXY OR SILICONE BASED, NON-CRACKING ELASTOMERIC SEALANT.

PAINTING

1. CONTRACTOR SHALL BE RESPONSIBLE FOR REPAINTING AREAS OF CONSTRUCTION THAT ARE SCRATCHED, MARRED, OR DAMAGED BY THE NEW CONSTRUCTION. CONTRACTOR SHALL MATCH THE COLOR, TYPE AND THICKNESS OF PAINT AS PREVIOUS.

ACCEPTANCE TESTING

1. UPON COMPLETION OF WORK, THE ENTIRE WIRING SYSTEM SHALL BE TESTED, AND SHALL BE SHOWN TO BE IN PROPER WORKING CONDITION IN ACCORDANCE WITH INTENT OF SPECIFICATIONS AND DRAWINGS. IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO HAVE ALL SYSTEMS READY FOR OPERATION AND TO HAVE AN ELECTRICIAN AVAILABLE TO OPERATE SAME IN ACCORDANCE WITH AND UNDER THE SUPERVISION OF THE INSPECTION REPRESENTATIVE OF THE ARCHITECT/ENGINEER. THE CONTRACTOR SHALL BE AVAILABLE TO ASSIST IN REMOVAL OF PANEL FRONTS, ETC., TO PERMIT INSPECTION AS REQUIRED.

THESE PLANS HAVE BEEN DESIGNED IN ACCORDANCE WITH THE FOLLOWING CODES AND REGULATIONS:

- 2020 NFPA 70 (NATIONAL ELECTRICAL CODE)

• 2023 FLORIDA FIRE PREVENTION CODE

• 2023 FLORIDA BUILDING CODE

• 2023 FBC – ENERGY CONSERVATION CODE

ELECTRICAL SYMBOLS LEGEND

POWER

-  20

ETR

DUPLEX RECEPTACLE, MOUNTED 18" AFF TO CENTERLINE UNLESS OTHERWISE NOTED. WP=WEATHERPROOF. NUMBER INDICATES BRANCH CIRCUIT. SEE PANEL BOARD SCHEDULE. ETR = EXISTING TO REMAIN.
- 

DUPLEX RECEPTACLE MOUNTED ABOVE COUNTER HEIGHT OR @ HEIGHT INDICATED. +44" AFF TO CENTER.
- 

GFI RECEPTACLE, MOUNTED 18" AFF TO CENTERLINE UNLESS OTHERWISE NOTED. WP DENOTES WEATHERPROOF COVER. NO FEED THRU TYPE DEVICES TO BE USED.
- 

GFI RECEPTACLE MOUNTED ABOVE COUNTER HEIGHT OR @ HEIGHT INDICATED. NO FEED THRU DEVICES TO BE USED. +44" AFF TO CENTER.
- 

C

JUNCTION BOX WITH IDENTIFIED BLANK COVER
C = CEILING MOUNTED
- 

TWO DUPLEX RECEPTACLES WITH COMMON COVER MOUNTED 18" AFF TO CENTERLINE UNLESS OTHERWISE NOTED.
- 

TWO DUPLEX GFI RECEPTACLES WITH COMMON COVER MOUNTED 18" AFF TO CENTERLINE UNLESS OTHERWISE NOTED.
- 

TWO DUPLEX GFI RECEPTACLES WITH COMMON COVER MOUNTED ABOVE COUNTER OR AT HEIGHT INDICATED. +44" AFF TO CENTER.
- 

TWO DUPLEX RECEPTACLES MOUNTED IN FLOOR BOX
- 

S0

SINGLE POLE SWITCH
(SUBSCRIPT LETTER INDICATES ITEM CONTROLLED)
- 

S3

THREE WAY SWITCH
(SUBSCRIPT LETTER INDICATES ITEM CONTROLLED)
- 

S0S

WALL MOUNTED DUAL TECHNOLOGY OCCUPANCY SENSOR SWITCH LITHONIA LIRW SERIES.
- 

S0S

CEILING MOUNTED DUAL TECHNOLOGY OCCUPANCY SENSOR LUTRON LOS-CDT SERIES. ** INDICATES FIXTURES IN SENSOR'S ZONE OF CONTROL.
- 

20A

SPECIAL PURPOSE RECEPTACLE. FIELD VERIFY EXACT NEMA CONFIGURATION. 20 AMP 240V.
- 

S30

2-POLE 30AMP MOTOR RATED SWITCH WITH INTEGRAL OVERLOADS.

SERVICE AND DISTRIBUTION

- 

BRANCH CIRCUIT PANELBOARD, UNDER 250 VOLTS, SURFACE MOUNTED
- 

30

FUSED DISCONNECT SWITCH. 3-POLE, 30AMP ASTERISK INDICATES TO FUSE ACCORDING TO MANUFACTURERS RECOMMENDATIONS. NEMA 1 RATED INDDOORS, NEMA 3R RATED OUTDOORS.
- 

30

NON-FUSED DISCONNECT SWITCH.3-POLE, 30AMP. NEMA 1 RATED INDDOORS, NEMA 3R RATED OUTDOORS.
- 

UTILITY TRANSFORMER
- 

TVSS

TRANSIENT VOLTAGE SURGE SUPPRESSOR. (TVSS)

LIGHTING (SEE LIGHTING FIXTURE SCHEDULE)

- 

2' X 4' LED LIGHT FIXTURE. NUMBER INDICATES BRANCH CIRCUIT WHICH FIXTURE IS CONNECTED. LOWER CASE LETTER INDICATES SWITCH WHICH CONTROLS FIXTURE. UPPER CASE LETTER INDICATES FIXTURE TYPE. SEE LIGHTING FIXTURE SCHEDULE. (NL = NIGHT LIGHT).
- 

INDUSTRIAL STRIP FLUORESCENT OR LED WITH TUBE AND WIRE GUARD
- 

RECESSED OR SURFACE MOUNTED COMPACT FLUORESCENT OR INCANDESCENT DOWNLIGHT. NUMBER INDICATES BRANCH CIRCUIT WHICH FIXTURE IS CONNECTED. LOWER CASE LETTER INDICATES SWITCH WHICH CONTROLS FIXTURE. UPPER CASE LETTER INDICATES FIXTURE TYPE. SEE LIGHTING FIXTURE SCHEDULE.
- 

WALL WASH / DIRECTIONAL FIXTURE
- 

WALL MOUNTED FLUORESCENT OR INCANDESCENT LIGHTING FIXTURE.
- 

RECESSED DECORATIVE CEILING MOUNTED ACCENT FIXTURE
- 

EMERGENCY EXIT SIGN SINGLE FACE WITH INTEGRAL 90 MINUTE EMERGENCY BATTERY POWER SUPPLY.
- 

EMERGENCY LIGHT WITH INTEGRAL BATTERY PACK – DUAL HEAD. C = CEILING MOUNT
- 

COMBINATION EMERGENCY LIGHT/EXIT SIGN SINGLE FACE WITH INTEGRAL 90 MINUTE EMERGENCY BATTERY POWER SUPPLY.
- 

SINGLE HEAD LED SITE LIGHTING FIXTURE
- 

DOUBLE HEAD LED SITE LIGHTING FIXTURE

DATA/COMM/TELEVISION

- 

DATA/COMM OUTLET WITH FLUSH MOUNTED 4-11/16" SQUARE SINGLE GANG JUNCTION BOX. PROVIDE BLANK BUILDING STANDARD COVER PLATE. EXTEND 1" CONCEALED CONDUIT TO ABOVE ACCESSIBLE CEILING AND STUB-OUT WITH PLASTIC BUSHING. OR AS NOTED. INSTALL PULL WIRE.
- 

TELEPHONE TERMINAL BOARD, 3/4" THICK x (LENGTH PER PLAN) x 8'-0" H. PLYWOOD BACKBOARD
- 

TELEVISION COAXIAL OUTLET WITH FLUSH MOUNTED 4-11/16" SQUARE SINGLE GANG JUNCTION BOX. PROVIDE BLANK BUILDING STANDARD COVER PLATE. EXTEND 1" CONCEALED CONDUIT TO ABOVE ACCESSIBLE CEILING AND STUB-OUT WITH PLASTIC BUSHING, OR AS NOTED. INSTALL PULL WIRE.
- 

COMBINATION POWER OUTLET AND DATA/COMM OUTLET MOUNTED FLUSH IN FLOOR. PROVIDE POWER CIRCUIT AND EMPTY 3/4" DATA/COMM CONCEALED CONDUIT TO VERTICAL WALL FOR ACCESS.

ABBREVIATIONS

- A – AMPERES

KAIC – (THOUSAND) AMPERE INTERRUPTING CAPACITY
- A/C – AIR CONDITIONING

KCMIL – THOUSANDS OF CIRCULAR MILLS
- AC – ALTERNATING CURRENT

KVA – KILOVOLT-AMPERES
- AFF – ABOVE FINISHED FLOOR

KW – KILOWATT
- AFG – ABOVE FINISHED GRADE

KW – KILOWATT
- AHU – AIR HANDLING UNIT

LED – LIGHT EMITTING DIODE
- AWG – AMERICAN WIRE GAUGE

LTG – LIGHTING
- C – CONDUIT

MCP – MOTOR CIRCUIT PROTECTION
- CFL – COMPACT FLUORESCENT LAMP

MCU – MASTER CONTROL UNIT
- CH – COUNTER HEIGHT

MH – METAL HALIDE
- CM – CARBON MONOXIDE

N – NEUTRAL
- CONC – CONCRETE

NA – NOT APPLICABLE
- COND – CONDUIT

NEC – NATIONAL ELECTRICAL CODE
- CU – COPPER

NEMA – NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
- DISC – DISCONNECT

NFPA – NATIONAL FIRE PROTECTION ASSOCIATION
- ECB – ENCLOSED CIRCUIT BREAKER

NF – NON-FUSED
- ELEC – ELECTRICAL

NIC – NOT IN CONTRACT
- EM – EMERGENCY

NL – NIGHT LIGHT
- EMS – ENERGY MANAGEMENT SYSTEM

O.C. – ON CENTER
- EMT – ELECTRICAL METALLIC TUBING

P – POLE
- EF – EXHAUST FAN

PNL – PANELBOARD
- EW – ELECTRIC WATER COOLER

PVC – POLYVINYL CHLORIDE
- EX – EXISTING

RGS – RIGID GALVANIZED STEEL
- FA – FIRE ALARM

RM – ROOM
- FACP – FIRE ALARM CONTROL PANEL

SPEC – SPECIFICATION
- FBC – FLORIDA BUILDING CODE

SS – STAINLESS STEEL
- GFI – GROUND FAULT INTERRUPTER

TVSS – TRANSIENT VOLTAGE SURGE SUPPRESSION
- GND – GROUND

TYP – TYPICAL
- GRC – GALVANIZED RIGID CONDUIT

UON – UNLESS OTHERWISE NOTED
- HVAC – HEATING, VENTILATING AND AIR CONDITIONING

V – VOLTS
- IWH – INSTANTANEOUS WATER HEATER

W – WIRE
- H.I.D. – HIGH INTENSITY DISCHARGE

WP – WEATHERPROOF
- I.G. – ISOLATED GROUND
- J – JUNCTION

LIGHTING FIXTURE SCHEDULE

TYPE	MANUFACTURER & CATALOG NO.	DESCRIPTION	LAMPS	VOLTS	OR APPROVED EQUALS BY
A	LITHONIA EPANL-2X4-4800LM-80CRI-40K-MVOLT	2'X4' LED RECESSED FLAT PANEL LENSED FIXTURE	INTEGRAL 45W LED	120	AS APPROVED BY OWNER
B	LITHONIA LDN6-40-40-L06-AR-LSS-MVOLT	6"ø LED RECESSED DOWNLIGHT	INTEGRAL 44W LED	120	AS APPROVED BY OWNER
C	TBD – SELECTED BY OWNER, FURNISHED AND INSTALLED BY CONTRACTOR	WALL MOUNTED LED VANITY LIGHT	PROVISION 50W LED	120	AS APPROVED BY OWNER
D	TBD – SELECTED BY OWNER, FURNISHED AND INSTALLED BY CONTRACTOR	LARGE DECORATIVE LED PENDANT	PROVISION 100W LED	120	AS APPROVED BY OWNER
E	EXITRONIX ILX-R-EM-WH	RED LED EMERGENCY EXIT SIGN WITH INTEGRAL 90 MINUTE BATTERY	INTEGRAL 5W LED	120	AS APPROVED BY OWNER
EC	EXITRONIX VLEDC-51-WH-G2	COMBINATION DUAL HEADS LED EMERGENCY LIGHTING AND EXIT SIGN WITH INTEGRAL 90 MINUTE BATTERY	INTEGRAL 5W LED	120	AS APPROVED BY OWNER
EM	EXITRONIX LED-95-WH-G2	WALL MOUNTED DUAL HEADS LED EMERGENCY LIGHTING FIXTURE WITH INTEGRAL 90 MINUTE BATTERY	INTEGRAL LED	120	AS APPROVED BY OWNER
F	TBD – SELECTED BY OWNER, FURNISHED AND INSTALLED BY CONTRACTOR	LED ROOPE LIGHTING MOUNTED IN COVE	PROVISION 50W LED	120	AS APPROVED BY OWNER
G	LITHONIA CLX L48 4000LM SEF FDL MVOLT GZ10 40K 80CRI	4'-0" LED LINEAR STRIP	INTEGRAL 28W LED	120	AS APPROVED BY OWNER

LIGHTING NOTES:

1. PROVIDE ALL DEVICES, MOUNTING, CLIPS, ESCUTCHEONS, PLATES AND SUPPORTS FOR ALL LIGHTING FIXTURES.
2. ALL LAMPING SHALL BE RATED 4000K FOR ALL LIGHTING FIXTURES, UON.
3. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN FOR EXACT FINAL LOCATION OF LIGHT FIXTURES.
4. COORDINATE THE ACTUAL TYPE OF CEILING FOR EACH FIXTURE WITH ARCHITECTURAL REFLECTED CEILING PLANS AND PROVIDE FIXTURE TRIM AS REQUIRED.
5. LAMPING CONFIGURATIONS ARE FOR THE DESIGN BASIS. ALTERNATE LAMPING CONFIGURATIONS ARE ACCEPTABLE, SUBJECT TO APPROVAL.



ARCHITECTS
3210 S. Osceola Ave.
Orlando, FL 32806
407-325-3550
www.eliteuniversalarchitects.com
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CONSULTANT

PROJECT

REVISIONS:

#	DATE	DESCRIPTION
02/16/26		SUBMIT

DRAWN BY: AFS

CHECKED BY: AFS

DATE: JANUARY 30, 2026

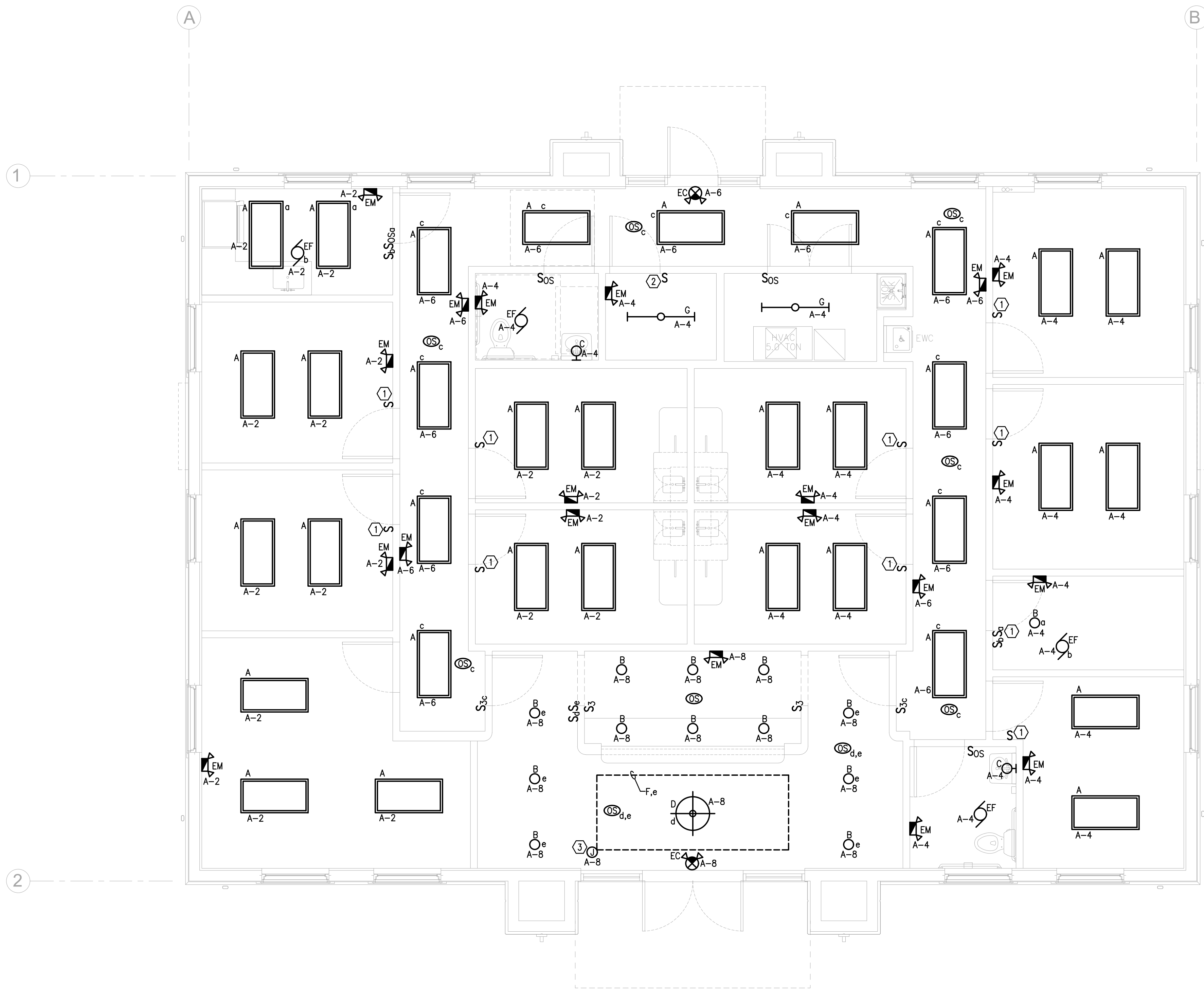
SHEET:

E001

Elite Universal Job No. 90210

ELECTRICAL SYMBOLS, ABBREV., NOTES & SCHEDULE

SOUTHERN RAILWAY BLDG #9 OAKLAND, FL 34787



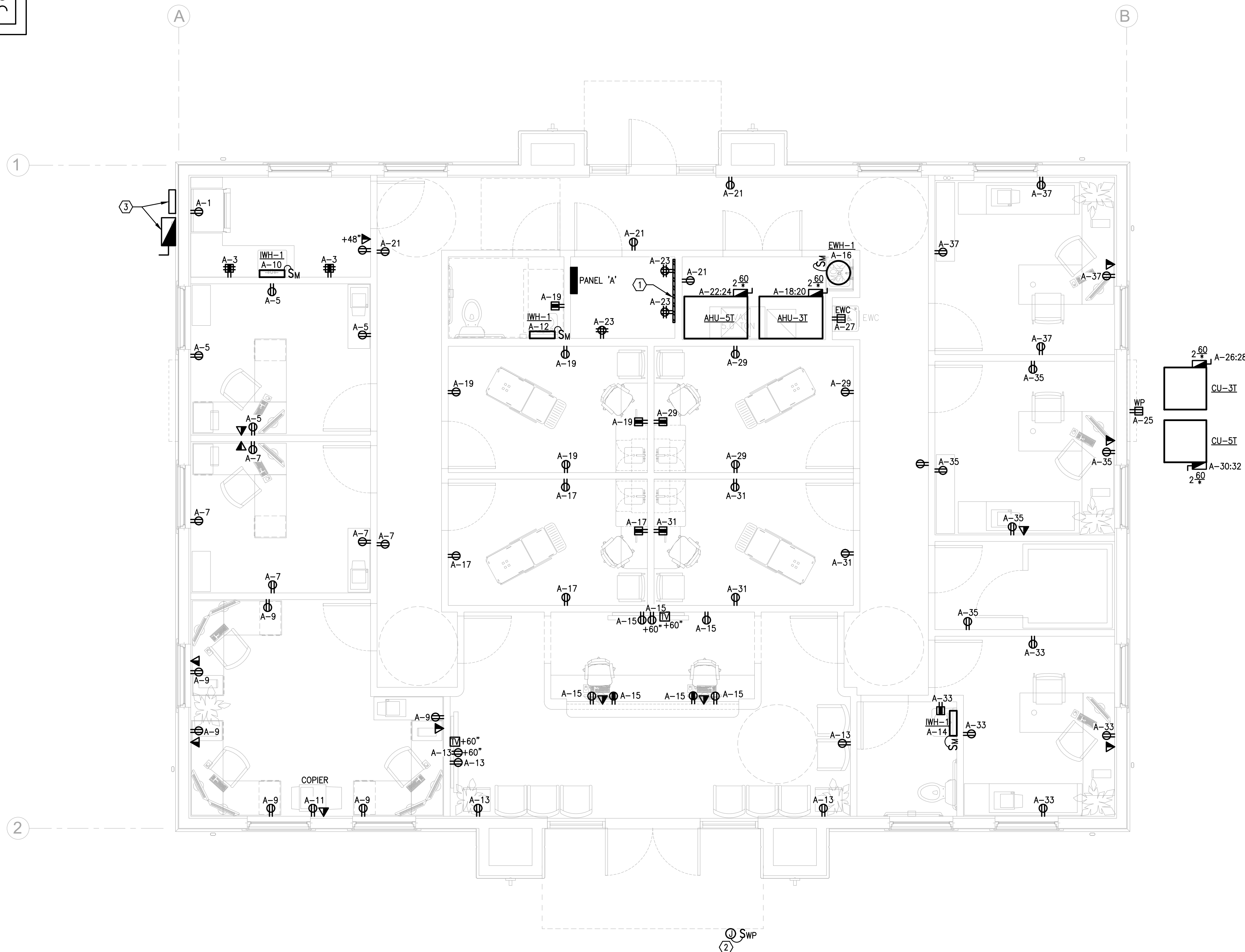
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E101
LIGHTING PLAN
SCALE: 1/4" = 1'-0"

GENERAL NOTES:

- ALL LIGHTING SHALL BE CONNECTED TO 120/208V 3Ø, 4W PANEL "A" UNLESS NOTED OTHERWISE.
- REFER TO ARCHITECTS REFLECTED CEILING PLAN FOR EXACT LIGHTING FIXTURE LOCATION.
- CONNECT ALL EMERGENCY LIGHTING AND EMERGENCY EXIT SIGNS AHEAD OF SWITCHING.
- ALL LIGHTING AND LIGHTING CONTROLS HAVE BEEN DESIGNED IN ACCORDANCE WITH 2023 FBC-ENERGY CONSERVATION SECTION C405.2.
- OCCUPANT SENSOR CONTROLS HAVE BEEN PROVIDED IN LOCATIONS, AND SHALL HAVE THE CONTROL FUNCTIONS, PER 2023 FBC-ENERGY CONSERVATION SECTION C405.2.1.
- LIGHTING MANUAL CONTROLS ARE LOCATED AND DESIGNED PER 2023 FBC-ENERGY CONSERVATION SECTION C405.2.6.
- DAYLIGHT-RESPONSIVE CONTROLS HAVE BEEN PROVIDED IN LOCATIONS AND HAVE CONTROL FUNCTIONS PER 2023 FBC-ENERGY CONSERVATION SECTION C405.2.4. PER EXCEPTION 1 THE AREA WHERE PATIENT CARE IS PROVIDED SHALL NOT BE REQUIRED TO HAVE DAYLIGHT-RESPONSIVE CONTROLS.
- THE TOTAL LOAD PER SIDE FOR ALL EXIT SIGNS DOES NOT EXCEED 5 WATTS, PER 2023 FBC-ENERGY CONSERVATION SECTION C405.4.1.
- INTERIOR LIGHTING POWER COMPLIES WITH 2023 FBC-ENERGY CONSERVATION SECTION C405.3.1. ALL LIGHTING FIXTURES IN THIS PROJECT'S SCOPE ARE ENERGY EFFICIENT LED FIXTURES.
- PER NEC 517.25, EMERGENCY LIGHTING HAS BEEN PROVIDED FOR ALL "PATIENT CARE" AREAS.
- PROVIDE THE REQUIRED FUNCTIONAL TESTING OF THE LIGHTING CONTROL SYSTEM PER FBC ENERGY CONSERVATION SECTION 408.3. PRIOR TO PASSING FINAL INSPECTION, THE DESIGN PROFESSIONAL SHALL PROVIDE EVIDENCE THAT THE LIGHTING CONTROL SYSTEMS HAVE BEEN TESTED TO ENSURE THAT CONTROL HARDWARE AND SOFTWARE ARE CALIBRATED, ADJUSTED, PROGRAMMED AND IN PROPER WORKING CONDITION IN ACCORDANCE WITH THE CONSTRUCTION DOCUMENTS AND MANUFACTURES INSTRUCTIONS. FUNCTIONAL TESTING SHALL BE IN ACCORDANCE WITH SECTIONS C408.3.1.1 AND C408.3.1.2 FOR APPLICATION CONTROL TYPE.
- THE DRAWINGS AND MANUALS SHALL BE SUBMITTED TO OWNER PER FBC ENERGY CONSERVATION SECTION C405.5.4.2.
- EXHAUST FANS SHALL BE INTERLOCKED WITH LOCAL LIGHTING CONTROL SWITCH.
- EXTERIOR LIGHTING FIXTURES SHALL BE CONTROLLED VIA INTEGRAL PHOTOCELL.

REFERENCE NOTES:

- OCCUPANT SENSOR CONTROLS SHALL NOT BE REQUIRED FOR THIS AREA PER 2023 FBC-ENERGY CONSERVATION C405.2.1.
- OCCUPANT SENSOR CONTROLS SHALL NOT BE REQUIRED FOR THIS AREA PER NEC 110.26.
- PROVIDE JUNCTION BOX AND 120V 20A BRANCH CIRCUIT MOUNTED IN CEILING FOR ROPE LIGHTING. CONTRACTOR SHALL COORDINATE LOCATION PRIOR TO ROUGH-IN.



1
E102

SCALE: 1/4" = 1'-0"

1. ALL POWER DEVICE SHALL BE CONNECTED TO 120/208V 3Ø, 4W PANEL "A", UNLESS NOTED OTHERWISE.
 2. COORDINATE WITH OWNER FOR VOICE/DATA REQUIREMENTS.
 3. THERE IS NO FIRE ALARM IN THIS PROJECT'S SCOPE.
 4. THIS PROJECT IS IN COMPLIANCE WITH 2023 FBC—ENERGY CONSERVATION C405.5.3 AND NEC 210.19 WITH REGARDS TO VOLTAGE DROP FOR BOTH FEEDER/CUSTOMER—OWNED SERVICE CONDUCTORS (2X) AND BRANCH CIRCUIT CONDUCTORS (3X).
 5. REFER TO ARCHITECTURAL MILLWORK ELEVATIONS FOR EXACT POWER RECEPTACLE LOCATION AND MOUNTING HEIGHTS.
 6. PER NEC 517.13, ALL BRANCH CIRCUITS SERVING ALL PATIENT CARE / EXAM ROOMS SHALL HAVE REDUNDANT GROUNDING BY ONE OF THE FOLLOWING METHODS:
 - INSTALLING FEEDER CONDUCTORS IN ELECTRICAL METALLIC TUBING (EMT).
 - INSTALLING FEEDER CONDUCTORS IN HOSPITAL GRADE MC CABLE (SOUTHWIRE HCF MCPAP OR EQUAL).
- "PATIENT CARE AREAS" ARE DEFINED PER NEC 517.2.
- HVAC UNIT LOCATIONS MAY VARY SLIGHTLY FROM MECHANICAL DRAWINGS FOR THE SAKE OF CLARITY. REFER TO MECHANICAL DRAWINGS FOR EXACT HVAC UNITS' LOCATIONS.
8. ALL WIRING SHALL BE MEDICAL GRADE.

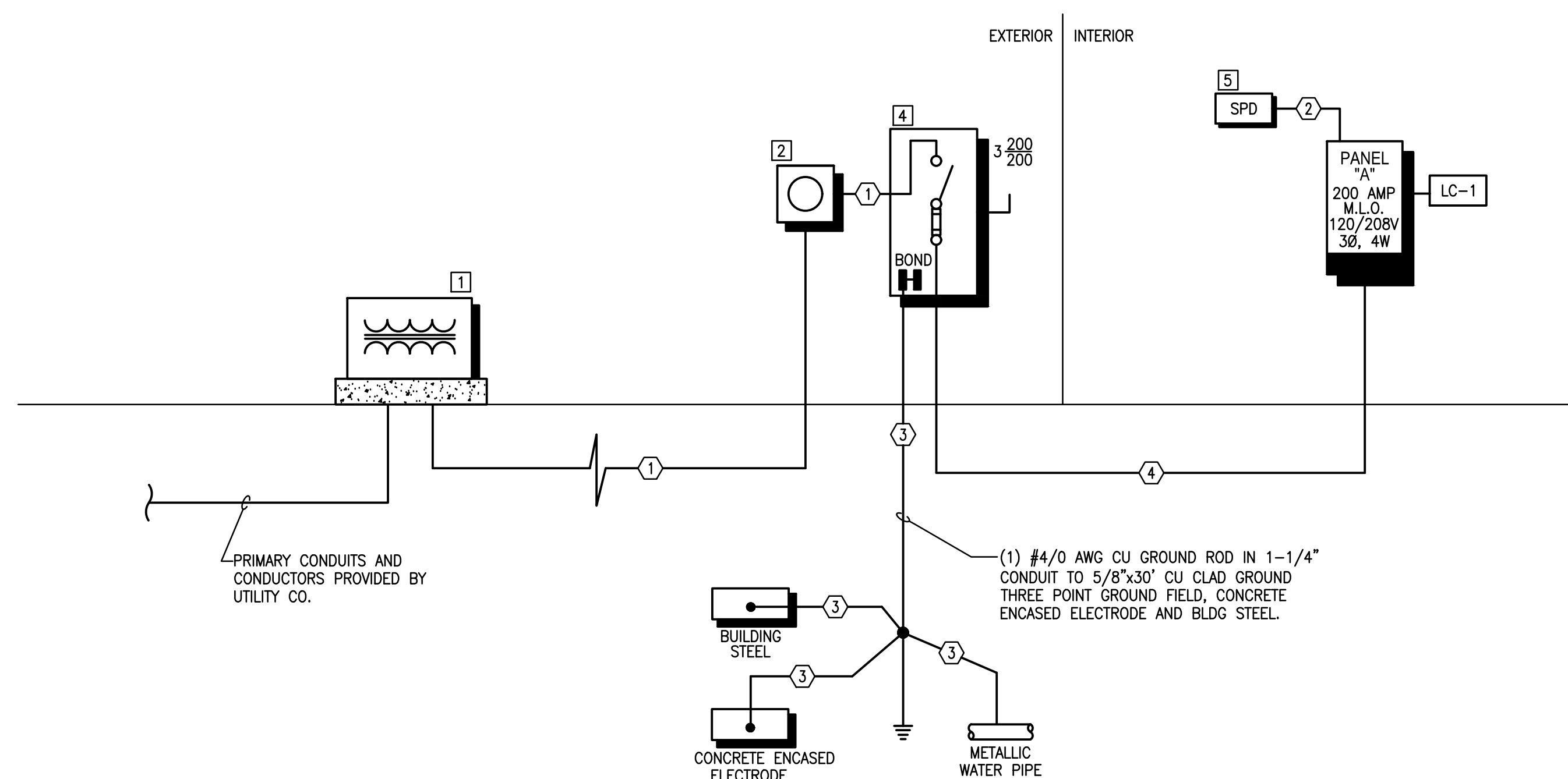
- ① **PLYWOOD BACKBOARD FOR PHONE/DATA/TV EQUIPMENT.**
PROVIDE 3/4" CDX PLYWOOD, UTILIZING 4'x8' SHEETS, WITH 2 COATS OF FIRE RETARDANT PAINT. INSTALL 6" FACE WITH TOP AT 8"-8" AFF. BACKBOARDS SHALL BE AC GRADE (FACE "A" SIDE INTO ROOM) FIRE RETARDANT PLYWOOD (HOOVER PYRO-GUARD). COUNTERSINK ALL SCREWS, FILL AND SMOOTH ALL SEAMS, COUNTERSINK SCREW HOLES AND VOIDS. ROUGH ALL ELECTRICAL OUTLETS INTO SPACE BEHIND BACKBOARD FOR FLUSH MOUNT INSTALLATION OF FACEPLATES.
- ② **PROVIDE WEATHERPROOF TOGGLE SWITCH, JUNCTION BOX AND 120V 20A BRANCH CIRCUIT A-39 FOR TENANT SIGNAGE.**
COORDINATE EXIST MOUNTING HEIGHT AND LOCATION PRIOR TO ROUGH-IN.
- ③ **LOCATION OF MAIN INCOMING ELECTRICAL SERVICE (DISCONNECT SWITCH AND METER). REFER TO ELECTRICAL RISER DIAGRAM FOR REQUIREMENTS.**

REVISIONS:	
DATE	DESCRIPTION
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DRAWN BY: AFS

CHECKED BY: AFS

DATE: JANUARY 30, 2026



ELECTRICAL RISER DIAGRAM

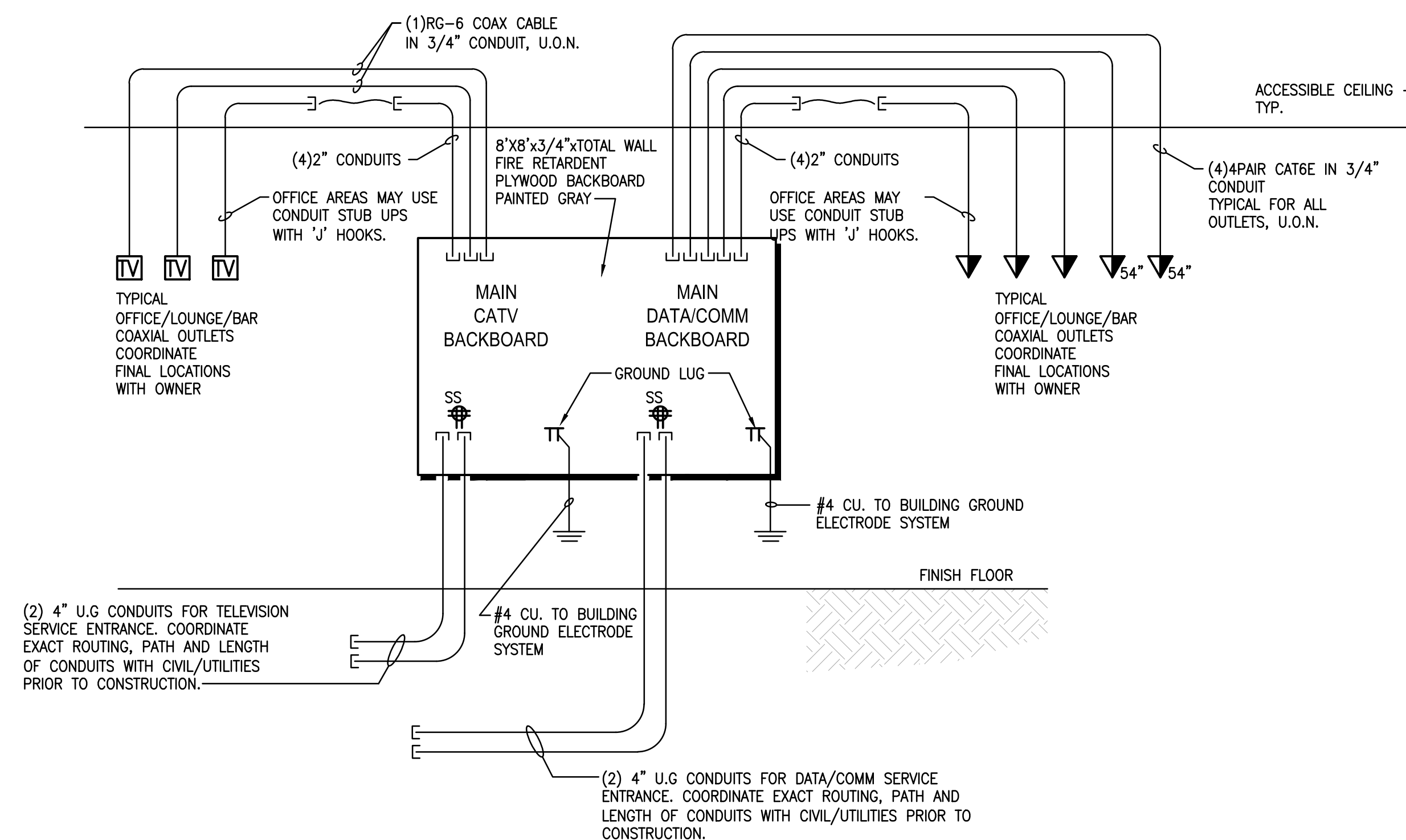
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ELECTRICAL RISER NOTES:

- 1 UTILITY COMPANY PAD MOUNTED UTILITY TRANSFORMER. PROVIDE CONCRETE HOUSEKEEPING PAD PER UTILITY COMPANY REQUIREMENTS. SECONDARY VOLTAGE = 120/208V 3ø, 4W.
- 2 EXISTING 120/208V 3ø, 4W, WP METER MOUNTED ON BUILDING. VERIFY IN FIELD "VIF"
- 3 EXISTING VIF WEATHERPROOF CURRENT TRANSFORMER CABINET.
- 4 120/208V 3ø, 4W, NEMA 3R, 200AMP SERVICE RATED FUSED DISCONNECT SWITCH WITH 200AMP 22KAIC FUSES.
- 5 SURGE PROTECTION DEVICE. TOTAL PROTECTION SOLUTIONS "SERVICE TRACK" TK-SERIES OR EQUAL.

WIRE AND CONDUIT SCHEDULE

SYMBOL	CONDUCTORS		COPPER GROUND	CONDUIT
	PER/PHASE	NEUTRAL		
	#3/0 AWG CU	#3/0 AWG CU	---	2"
	#10 AWG CU	#10 AWG CU	#10 AWG CU	3/4"
	---	---	#4/0 AWG CU	1-1/4"
	#3/0 AWG CU	#3/0 AWG CU	#6 AWG CU	2"



DATA/T.V./COMMUNICATION RISER DIAGRAM

NTS

SEE PROJECT SPECIFICATIONS FOR CONDUIT AND WIRE REQUIREMENTS.

42 CIRCUITS 128 AMPS

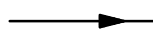
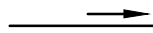
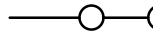
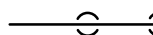
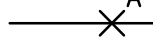
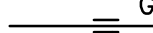
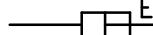
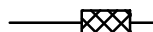
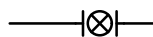
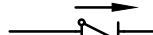
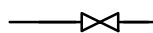
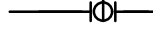
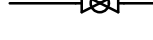
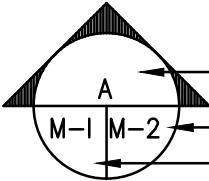
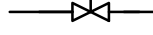
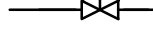
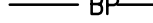
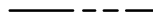
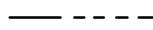
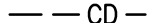
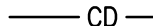
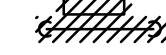
* = AHU AND CU LOADS ARE NON-COINCIDENTAL. ONLY THE LARGEST OF THE TWO LOADS IS USED IN THE LOAD CALCULATIONS.
NOTE: ELECTRICAL CONTRACTOR SHALL VERIFY HVAC BREAKERS/FUSES ACTUAL SIZES WITH MECHANICAL AND MANUFACTURER CUT SHEET/SUBMITTALS & REFER TO INSTALLATION MANUAL PRIOR TO BIDDING THE PROJECT



SPECIFICATION	
PART 1 GENERAL 1.1 SECTION INCLUDES A. GENERAL PROVISIONS SPECIFICALLY APPLICABLE TO DIVISION 15 SECTIONS, IN ADDITION TO DIVISION 1 – GENERAL REQUIREMENTS. 1.2 SCOPE A. THE WORK SHALL INCLUDE THE PROVISIONS OF SYSTEMS, EQUIPMENT AND MATERIALS SPECIFIED IN THIS DIVISION AND AS CALLED FOR ON THE DRAWINGS. WORK SHALL ALSO INCLUDE SUPERVISION, OPERATION, METHODS AND LABOR FOR THE FABRICATION, START-UP AND TESTS FOR A COMPLETE OPERATIONAL PLUMBING INSTALLATION. B. DRAWINGS FOR THE WORK ARE DIAGRAMMATIC IN NATURE AND ARE INTENDED TO CONVEY THE SCOPE OF THE INSTALLATION AND TO INDICATE THE GENERAL ARRANGEMENT AND LOCATIONS OF THE WORK. BECAUSE OF THE SCALE OF THE DRAWINGS, CERTAIN BASIC ITEMS SUCH AS PIPE FITTINGS, ACCESS PANELS, AND SLEEVES MAY NOT BE SHOWN. THIS CONTRACTOR SHALL BE RESPONSIBLE FOR SELECTING THE EQUIPMENT TO FIT THE SPACE PROVIDED, THE LOCATION AND SIZES FOR PIPE FITTINGS, SLEEVES, INSERTS, FIRE AND/OR SMOKE DAMPERS, AND OTHER BASIC ITEMS REQUIRED BY CODE AND OTHER SECTIONS SHALL BE COORDINATED AND INCLUDED FOR THE PROPER INSTALLATION OF THE WORK. C. EQUIPMENT SPECIFICATIONS MAY NOT DEAL INDIVIDUALLY WITH MINUTE ITEMS REQUIRED SUCH AS COMPONENTS, PARTS, CONTROLS AND DEVICES WHICH MAY BE REQUIRED TO PRODUCE THE EQUIPMENT PERFORMANCE SPECIFIED OR AS REQUIRED TO MEET THE EQUIPMENT WARRANTIES. WHERE SUCH ITEMS ARE REQUIRED, THEY SHALL BE INCLUDED BY THE INSTALLER OF THE EQUIPMENT, WHETHER OR NOT SPECIFICALLY INDICATED IN THE CONTRACT DOCUMENTS WITH NO ADDITIONAL COST INCURRED. D. WHERE NOTED ON THE DRAWINGS OR INDICATED IN OTHER SECTIONS OF THE SPECIFICATION, THE CONTRACTOR FOR THIS DIVISION SHALL INSTALL PLUMBING EQUIPMENT FURNISHED BY OTHERS, AND SHALL MAKE ALL FINAL CONNECTIONS. CONTRACTOR SHALL INSTALL EQUIPMENT AND SYSTEMS IN STRICT ACCORDANCE WITH MANUFACTURERS RECOMMENDATIONS. E. COORDINATE WITH ALL TRADES IN SUBMITTAL OF SHOP DRAWINGS. TIGHT SPACE CONDITIONS SHALL BE DETAILED TO THE SATISFACTION OF ALL TRADES, SUBJECT TO THE REVIEW AND FINAL ACCEPTANCE OF THE ARCHITECT/ENGINEER. IN THE EVENT THAT THE CONTRACTOR INSTALLS HIS WORK BEFORE COORDINATING WITH OTHER TRADES OR SO AS TO CAUSE ANY INTERFERENCE, WITH THE WORK OF THE OTHER TRADES THIS CONTRACTOR SHALL MAKE ALL REQUIRED CHANGES TO CORRECT THE CONDITION AT NO ADDITIONAL COST TO THE PROJECT. F. CONTRACTOR SHALL VERIFY ALL EQUIPMENT CONNECTION SIZES PRIOR TO INSTALLATION OF ANY SYSTEMS. THIS CONTRACTOR SHALL ADJUST PIPING SYSTEM SIZES AS REQUIRED TO MATCH EQUIPMENT CONNECTIONS. UTILIZE REDUCERS WHERE EQUIPMENT CONNECTIONS ARE SMALLER THAN PIPE SIZES INDICATED ON PLANS; NO PIPING SHALL BE DECREASED IN SIZE (IN THE DIRECTION OF FLOW). G. EXAMINATION OF BIDDING DOCUMENTS: 1. EACH BIDDER SHALL EXAMINE THE BIDDING DOCUMENTS CAREFULLY AND NOT LATER THAN TEN (10) DAYS PRIOR TO THE DATE FOR RECEIPT OF BIDS, SHALL MAKE WRITTEN REQUEST TO THE ENGINEER FOR INTERPRETATION OR CORRECTION OF ANY AMBIGUITY, INCONSISTENCY OR ERROR THEREIN. ANY INTERPRETATION OR CORRECTION WILL BE ISSUED AS AN ADDENDUM BY THE ENGINEER. ONLY A WRITTEN INTERPRETATION OR CORRECTION TO BID DOCUMENTS BY ADDENDUM SHALL BE BINDING. NO BIDDER SHALL RELY UPON ANY INTERPRETATION OR CORRECTION GIVEN BY ANY OTHER METHOD. H. SUBSTITUTIONS 1. EXCEPT AS PROVIDED BELOW, EACH BIDDER REPRESENTS THAT HIS BID IS BASED UPON THE MATERIALS AND EQUIPMENT DESCRIBED IN THE BIDDING DOCUMENTS. 2. NO SUBSTITUTIONS FOR OTHER MATERIALS AND EQUIPMENT WILL BE CONSIDERED UNLESS WRITTEN REQUEST HAS BEEN SUBMITTED TO THE ENGINEER FOR APPROVAL AT LEAST TEN (10) DAYS PRIOR TO THE DATE FOR RECEIPT OF BIDS. EACH SUCH REQUEST SHALL INCLUDE A COMPLETE DESCRIPTION OF THE PROPOSED SUBSTITUTE, THE NAME OF THE MATERIAL OR EQUIPMENT FOR WHICH IT IS TO BE SUBSTITUTED, DRAWINGS, CUTS, PERFORMANCE AND TEST DATA OR INFORMATION NECESSARY FOR A COMPLETE EVALUATION. 3. IF THE ENGINEER APPROVES ANY PROPOSED SUBSTITUTION, SUCH APPROVAL WILL BE ISSUED AS AN ADDENDUM FORWARDED TO ALL PARTICIPATING BIDDERS. 4. IF ANY BIDDER IS UNABLE TO PROCURE WRITTEN APPROVAL OF ANY SUBSTITUTION FROM THE ENGINEER PRIOR TO THE OPENING OF BIDS, THEN HE SHALL BASE HIS BID ON ITEMS AND SYSTEMS AS SPECIFIED IN BID DOCUMENTS. 5. SUBSTITUTIONS REQUESTED ON THE BID PROPOSAL FORM WHICH ARE PRIOR APPROVED BY THE ENGINEER WILL BE INCORPORATED INTO THE CONTRACT WITH THE SUCCESSFUL BIDDER. 6. REQUESTS FOR ANY SUBSTITUTIONS NOT SUBMITTED AND APPROVED IN ACCORDANCE WITH THE ABOVE INSTRUCTIONS WILL BE DENIED BY THE ENGINEER. 1.3 REGULATORY REQUIREMENTS A. ALL INSTALLATIONS AND EQUIPMENT SHALL BE IN STRICT ACCORDANCE WITH ALL APPLICABLE STATUTES, CODES AND REGULATIONS OF THE GOVERNMENTAL BODIES HAVING JURISDICTION. B. CONFORM TO 2020–7TH ED. FLORIDA BUILDING CODE. C. CONFORM TO 2018 FLORIDA FIRE PREVENTION CODE. D. NATIONAL ELECTRIC CODE 2017 EDITION. 1.4 CUTTING AND PATCHING A. SUBMIT WRITTEN REQUEST IN ADVANCE OF CUTTING OR ALTERING ELEMENTS. B. EMPLOY SKILLED AND EXPERIENCED INSTALLER TO PERFORM CUTTING AND PATCHING WHICH EFFECT: 1. STRUCTURAL INTEGRITY OF ELEMENT. 2. INTEGRITY OF WEATHER. 3. EFFICIENCY, MAINTENANCE, OR SAFETY OF ELEMENT. 4. VISUAL QUALITIES OF SIGHT. 5. WORK OF OWNER OR SEPERATE CONTRACTOR.	PART 2 PRODUCTS 2.1 PIPE HANGERS AND SUPPORTS A. MANUFACTURERS: 1. B–LINE. 2. OTHER ACCEPTABLE MANUFACTURERS OFFERING EQUIVALENT PRODUCTS. A) MICHIGAN HANGER. B) PHD. 2.2 HOT WATER WATER PIPING INSULATION A. MANUFACTURERS: 1. OWENS–CORNING 2. JOHNS–MANVILLE 3. KNAUF B. ASTM C547, RIGID MOLDED NON–COMBUSTIBLE FIBERGLASS INSULATION. K' VALUE .024 @ 75 DEGREES F. C. VAPOR BARRIER JACKET, KRAFT PAPER WITH GLASS FIBER YARN AND BONDED TO ALUMINUM FILM. CLOSURE SYSTEM SHALL BE UL 181. D. INSULATION THICKNESS 1–1/2" 2.3 PIPING: STORM A. SOLID–WALL SCHEDULE 40 PVC PIPE: ASTM D 2665 1. PVC SOCKET FITTINGS: ASTM D 2665, MADE TO ASTM D 3311, DRAIN, WASTE, AND VENT PATTERNS AND TO FIT SCHEDULE 40 PIPE. 2.5 2.4 PIPING: SANITARY SEWER PIPING, ABOVE AND BELOW GRADE A. PVC: ASTM D2665 SCHEDULE 40 PVC PIPING. 1. FITTINGS: ASTM D2665 SCHEDULE 40 PVC DWV FITTINGS. 2. JOINTS: ASTM D2564 SOLVENT CEMENT. JOINTS SHALL BE INSTALLED IN ACCORDANCE WITH ASTM D2855. 2.5 PIPING: WATER PIPING, ABOVE GRADE A. COPPER PIPING: ASTM B88 TYPE 'L' HARD DRAWN. 1. FITTINGS: ASME 16.18 CAST BRONZE OR ASME B16.22 WROUGHT COPPER OR BRONZE. 2. JOINTS: ASTM B32 SOLDER, ASTM 95–5 TA MINIMUM, FLUX ASTM B813. JOINTS SHALL BE INSTALLED IN ACCORDANCE WITH ASTM B828. B. CPVC
	PART 3 INSTALLATION A. INSTALL IN STRICT ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. B. PROVIDE NON–CONDUCTING DIELECTRIC CONNECTIONS WHEREVER JOINTING DISSIMILAR METALS. C. ROUTE PIPING IN ORDERLY MANNER AND MAINTAIN GRADIENT. D. INSTALL PIPING TO CONSERVE BUILDING SPACE AND NOT INTERFERE WITH USE OF SPACE. E. GROUP PIPING WHENEVER PRACTICAL AT COMMON ELEVATIONS. F. INSTALL PIPING TO ALLOW FOR EXPANSION AND CONTRACTION WITHOUT STRESSING PIPE, JOINTS, OR CONNECTED EQUIPMENT. G. PROVIDE CLEARANCE FOR INSTALLATION OF INSULATION AND ACCESS TO VALVES AND FITTINGS. H. PROVIDE ACCESS WHERE VALVES AND FITTINGS ARE NOT EXPOSED. COORDINATE SIZE AND LOCATION OF ACCESS DOORS WITH APPROPRIATE ARCHITECTURAL SECTION. I. ESTABLISH ELEVATIONS OF BURIED PIPING OUTSIDE THE BUILDING TO ENSURE PROPER SLOPE AND COVER. J. ESTABLISH ELEVATIONS OF EXISTING BURIED PIPING FOR CONNECTION TO NEW WORK TO ENSURE PROPER SLOPE AND COVER. K. WHERE PIPE SUPPORT MEMBERS ARE WELDED TO STRUCTURAL BUILDING FRAMING, SCRAPE, BRUSH CLEAN, AND APPLY ONE COAT OF ZINC RICH PRIMER TO WELDING. L. EXTEND CLEANOUTS TO FINISHED FLOOR OR WALL SURFACE. LUBRICATE THREADED CLEANOUT PLUGS WITH MIXTURE OF GRAPHITE AND LINED OIL. ENSURE CLEARANCE AT CLEANOUT FOR RODDING OF DRAINAGE SYSTEM. M. RESET SPRINKLER HEADS FROM UPRIGHT POSITION TO DOWN POSITION TO CENTER OF NEW CEILING TILE. K. TORQUE ALL HUBLESS CAST IRON COUPLING IN STRICT ACCORDANCE WITH THE MANUFACTURERS RECOMMENDATIONS. UTILIZE COUPLING MANUFACTURERS PRE–SET TORQUE WRENCH. O. INSTALL WATER HAMMER SHOCK ARRESTORS AT EACH FIXTURE OR BATTERY OF FIXTURES WHERE REQUIRED. ARRESTORS SHALL BE FACTORY FABRICATED. INSTALL ARRESTORS AND SIZE PER PLUMBING AND DRAINAGE INSTITUTE STANDARD P.D.I. WH–201. ACCEPTABLE MANUFACTURERS ZURN, JOSAM, SIOUX CHIEF. PPP INC. FOR 1–11 FU'S PROVIDE PDI–A, FOR 12–32 FU'S PDI–B, 33–60 FU'S PDI–C, 61–113 FU'S PDI–D.
	NOTES TO CONTRACTOR 1. GATE VALVES SHALL NOT BE USED ON THIS JOB. ONLY BALL VALVES WILL BE APPROVED AND THEY SHALL BE "NIBCO" #S–585 FOR SI'1EAT SOLDER OR "NIBCO" # T–585 FOR THREADED.' BOTH ARE TESTED FOR 600 P51 W/FULL PORT DIA. WHICH WILL NOT RESTRICT FLOW 2. ALL PLUMBING VALVES, LINES AND EQUIPMENT SHALL BE PROPERLY LABELED 3. PLUMBING CONTRACTOR (PC) SHALL SUPPLY & INSTALL MIXING VALVES (SYMMONS NO. 4–10B) ON ALL KITCHEN HAND SINKS & LAVATOIRES FOR 110 DEG. F. MAX. HOT WATER SUPPLY. 4. PLUMBING CONTRACTOR SHALL SUPPLY & INSTALL CHECK VALVES ON ALL BAR SINK SWING FAUCETS, SERVICE SINK, PRE–RINSE UNIT SUPPLIES TP PREVENT BACKFLOW 5. INSULATE ALL COLD AND HOT WATER PIPING 6. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE ADEQUATE SUPPORT FOR THE EXPANSION TANK. SUPPORT TO ADJACENT WALL OR FROM EXISTING STRUCTURE ABOVE WITH UNISTRUT. DO NOT SUPPORT FROM PIPE ALONE. PROVIDE DRAIN PAN UNDER WATER HEATER. PROVIDE SEISMIC STRAPPING ON WATER HEATER ABOVE MOB SINK. TERMINATE (EWH–2) DRAIN LINE IN JANITOR SINK (JS–1). PROVIDE FULL SIZE T&P REF. LINE. REFER TO WATER HEATER DETAIL ON SHEET P–5.1 FOR ADDITIONAL INFORMATION.

ABBREVIATIONS	
A	AMPERES
ACS	AUTOMATIC CONTROL SYSTEM
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
APPROX	APPROXIMATELY
AP	ACCESS PANEL
ARCH	ARCHITECTURAL
AUTO	AUTOMATIC
AUX	AUXILIARY
BHP	BRAKE HORSEPOWER
BLDG	BUILDING
BTU	BRITISH THERMAL UNIT
BTUH	BRITISH THERMAL UNITS PER HOUR
BWV	BACK WATER VALVE
CCP	CENTRAL CONTROL PANEL
CD	CONDENSATE DRAIN
CLG	CEILING
CO	CLEAN-OUT
COMB	COMBINATION
COMPR	COMPRESSOR
COND	CONDENSATE OR CONDENSER
CONT	CONTINUATION
CMU	CONCRETE MASONRY UNIT
CU FT	CUBIC FEET
CU IN	CUBIC INCHES
CW	COLD WATER (CITY)
D	DRAIN LINE
EL	ELEVATION
ELEC	ELECTRIC
EQ	EQUAL
EXST	EXISTING
EXP	EXPANSION
EO	EMERGENCY OVERFLOW PIPING
ESEW	EMERGENCY SHOWER/EYEWASH
'F	DEGREES FAHRENHEIT
ECO	EXTERIOR CLEANOUT
FC	FLEXIBLE CONNECTION
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
FG	FINISHED GRADE
FIN FL	FINISHED FLOOR
FLA	FULL LOAD AMPERES
FOB	FLAT ON BOTTOM
FOT	FLAT ON TOP
FBM	FEET PER MINUTE
FPS	FEET PER SECOND
GA	GAUGE
GAL	GALLONS
GPH	GALLONS PER HOUR
GO	GAS OUTLET
GPM	GALLONS PER MINUTE
H	HEIGHT
HB	HOSE BIBB
H2O	WATER
HD	HEAD
HORIZ	HORIZONTAL
HP	HORSEPOWER
HW	HOT WATER
HR	HOUR
HZ	FREQUENCY
ID	INSIDE DIAMETER
IN	INCH OR INCHES
INSUL	INSULATION
IPS	IRON PIPE SIZE
INV	INVERT ELEVATION
KW	KILOWATT
L	LENGTH
LB/HR	POUNDS PER HOUR
LBS	POUNDS
LIN FT	LINEAR FEET
MAX	MAXIMUM
MBH	THOUSANDS, BTUH
MC	MECHANICAL CONTRACTOR
MIN	MINIMUM
NC	NORMALLY CLOSED
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN
NO.	NUMBER
NTS	NOT TO SCALE
OD	OUTSIDE DIAMETER
OV	OUTLET VELOCITY
P	PRESSURE
PC	PLUMBING CONTRACTOR
PD	PRESSURE DROP
PSI	POUNDS PER SQUARE INCH
PSIA	PSI ABSOLUTE
PSIG	PSI GAUGE
PRESS	PRESSURE
PVC	POLYVINYL CHLORIDE
R	RISE
RD	ROOF DRAIN
REQ'D	REQUIRED
RM	ROOM
RPM	REVOLUTIONS PER MINUTE
RV	RELIEF VALVE
SAN	SANITARY
SMH	STORM MANHOLE
SPEC	SPECIFICATION
SCW	SOFT COLD WATER
ST	STORM
TD	TRENCH DRAIN
TDH	TOTAL DYNAMIC HEAD
TEMP	TEMPERATURE
TYP	TYPICAL
V	VENT LINE
VLV	VALVE
VTR	VENT THRU ROOF
W	WIDTH
W/	WITH
W/O	WITHOUT
WC	WATER COLUMN
WCO	WALL CLEANOUT
WG	WATER GAUGE
WP	WORKING PRESSURE

PLUMBING LEGEND

	PIPE SECTION—SUPPLY		— ST —	STORM SEWER (BELOW GRADE)
	PIPE SECTION—RETURN		— ST —	STORM SEWER (ABOVE GRADE)
	DIRECTION OF FLOW IN PIPE		— EO —	STORM EMERGENCY OVERFLOW PIPING
	PITCH PIPE DOWN IN DIRECTION OF ARROW		— GR —	GREASE SEWER (BELOW GRADE)
	PIPE UP		— GR —	GREASE SEWER (ABOVE GRADE)
	PIPE DOWN		— AW —	ACID WASTE (BELOW GRADE)
	PIPE ANCHOR		— AW —	ACID WASTE (ABOVE GRADE)
	PIPE GUIDE		— AV —	ACID WENT
	EXPANSION JOINT		— SCW —	SOFT COLD WATER
	FLEXIBLE PIPE CONNECTOR		— — —	VENT
	BALL VALVE		— G —	GAS LINE (BELOW GRADE)
	CHECK VALVE, HORIZONTAL SWING		— G —	GAS LINE (ABOVE GRADE)
	GATE VALVE			POINT OF CONNECTION NEW TO EXISTING
	BUTTERFLY VALVE			POINT OF REMOVAL FOR DEMOLITION
	GLOBE VALVE		ΔT	DELTA T (TEMPERATURE DIFFERENCE)
	BALANCING VALVE			SECTION DESIGNATION
	BALANCING COCK			SHEET WHERE SECTION APPEARS SHEET WHERE SECTION CUT
	STRAINER, Y-TYPE AND BLOWOFF VALVE		HB	HOSE BIBB
	PRESSURE RELIEF VALVE (WATER)		FD	FLOOR DRAIN
	PRESSURE REDUCING VALVE		HD	HUB DRAIN
	MANUAL AIR VENT		TD	TRENCH DRAIN
	SOLENOID VALVE		— A —	COMPRESSED AIR (BELOW GRADE)
	VALVE OS & Y		— A —	COMPRESSED AIR (ABOVE GRADE)
	CAPPED LINE		— HOR —	HEATING OIL RETURN (BELOW GRADE)
	SANITARY SEWER (ABOVE GRADE)		— HOR —	HEATING OIL RETURN (ABOVE GRADE)
	SANITARY SEWER (BELOW GRADE)		— HOS —	HEATING OIL SUPPLY (BELOW GRADE)
	DOMESTIC COLD WATER		— HOS —	HEATING OIL SUPPLY (ABOVE GRADE)
	BOOSTED PRESSURE DOMESTIC COLD WATER		— — —	FLOOR CLEANOUT
	DOMESTIC HOT WATER		— — —	CLEANOUT
	DOMESTIC HOT WATER RETURN		— — —	EXTERIOR CLEANOUT
	CONDENSATE DRAIN (BELOW GRADE)		— — —	HORIZONTAL CLEANOUT
	CONDENSATE DRAIN (ABOVE GRADE)			ITEMS TO BE REMOVED (DEMOLITION)
			— CWN —	COLD NANO WATER
			— CWF —	COLD FILTERED WATER
				SHOCK ABSORBER

PLUMBING GENERAL NOTES	
1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN STRICT ACCORDANCE WITH APPLICABLE LOCAL CODES, RULES AND ORDINANCES.	18. PROVIDE CHROME-PLATED COMBINATION COVER PLATE AND CLEANOUT PLUG FOR ALL WALL CLEANOUTS ZURN - ZN-1446 OR EQUAL.
2. THE CONTRACTOR SHALL VISIT THE JOB SITE AND THOROUGHLY FAMILIARIZE HIMSELF WITH ALL EXISTING CONDITIONS.	19. PROVIDE EACH FIXTURE GROUP WITH ISOLATION VALVES, BOTH HOT (110) AND COLD WATER.
3. ALL WORK SHALL BE PERFORMED BY A LICENSED PLUMBING CONTRACTOR, IN A FIRST-CLASS AND WORKMANLIKE MANNER. THE COMPLETE SYSTEM SHALL BE FULLY OPERATIVE.	20. NO COMBUSTIBLE MATERIALS CAN BE USED IN MECHANICAL ROOMS OR IN CEILING SPACES WHERE USED AS RETURN AIR PLenums.
4. ALL EXCAVATION AND BACKFILL, AS REQUIRED, FOR THIS PHASE OF CONSTRUCTION SHALL BE A PART OF THIS CONTRACT.	21. PROVIDE BACKFLOW PREVENTER - WILKINS model # 575 OR EQUAL.
5. PROOF OF INSURANCE SHALL BE PROVIDED BY THE CONTRACTOR FOR PROTECTION AGAINST PUBLIC LIABILITY AND PROPERTY DAMAGE FOR THE DURATION OF THE WORK.	22. THE DRAWINGS ARE GENERALLY DIAGRAMMATIC AND INDICATE THE APPROXIMATE ROUTING OF PIPING AND LOCATION OF FIXTURES. THE CONTRACTOR SHALL COORDINATE WORK WITH OTHER TRADES AND MAKE MINOR OFFSETS AND ADJUSTMENTS AS REQUIRED AT NO ADDITIONAL COST TO THE OWNER.
6. VERIFY LOCATION, SIZE, INVERTS AND ALL EXISTING UTILITIES PRIOR TO BEGINNING OF CONSTRUCTION. ADVISE ENGINEER OF ANY DISCREPANCIES.	23. COORDINATE FIXTURES LOCATIONS WITH ARCHITECTURAL DRAWINGS.
7. WATER PIPING SHALL BE TYPE "L" COPPER FOR 2 1/2" ABOVE GRADE. ALL UNDERGROUND WATER PIPING SHALL BE TYPE 2K3 COPPER WITH NO JOINTS UNDER SLAB.	24. CONTRACTOR SHALL MAKE ARRANGEMENTS FOR CONNECTIONS TO ALL UTILITY LINES AND PAY ALL FEES AND COSTS FOR CONNECTIONS TO THOSE SERVICES.
8. SOIL, WASTE, VENT AND RAINWATER PIPING SHALL BE CAST IRON NO. HUB 301-72 ABOVE GRADE. MECHANICAL JOINTS WITH S.S. CLAMPS BELOW GRADE. (USE SCHEDULE 40 PVC WHERE ACCEPTABLE BY CODE, AT THE CONTRACTORS OPTION.)	25. ALL PIPING SHALL BE RUN IN CONCEALED LOCATIONS EXCEPT WHERE NOTED.
9. AIR CONDITIONING CONDENSATE DRAIN PIPING SHALL BE COPPER DWV PIPE AND FITTINGS. INSULATE ALL CONDENSATE PIPING ABOVE GRADE.	26. PLUMBING FIXTURES SHALL BE FIRST QUALITY VITREOUS CHINA, STAINLESS STEEL OR PLASTIC AS NOTED ON FIXTURE SCHEDULE. ALL FIXTURES SHALL BE RIGIDLY CONNECTED TO THE BUILDING AND SHALL BE CLEANED AND FUNCTIONAL PRIOR TO ACCEPTANCE.
10. INSULATE ALL HOT WATER SUPPLY, HOT WATER RETURN, RAINWATER AND CONDENSATE LINES ABOVE GRADE AS FOLLOWS: (ONLY WHERE REQUIRED BY CODE.) HOT WATER SUPPLY AND RETURN: 1 1/2" THICK FIBERGLASS. RAINWATER LEADERS 1 1/2" THICK FIBERGLASS BLANKET ON DRAIN BODY AND 1" HORIZONTAL A/WL	27. REFER TO ARCHITECTURAL DRAWINGS FOR FINISHED GRADES.
11. ALL FIXTURES MUST BE PROVIDED WITH READILY ACCESSIBLE STOPS AND MARKED ACCESS PANELS.	28. CONTRACTOR SHALL PROVIDE PRESSURE REDUCING VALVE AND REDUCED PRESSURE BACKFLOW PREVENTION VALVE INSIDE BUILDING WHERE SERVICE ENTERS OR AS SHOWN ON THE SITE PLAN.
12. FURNISH AND INSTALL APPROVED AIR CHAMBERS AT EACH PLUMBING FIXTURE GROUP AND P.D. I. APPROVED SHOCK ARRESTERS ON MAIN LINES OR RISER.	29. EXPOSED PIPING BELOW FIXTURES SHALL BE CHROME PLATED. PIPING AT FIXTURES IN HANDICAPPED ACCESSIBLE AREAS SHALL BE INSULATED TO PROTECT AGAINST BURNS.
13. DIELECTRIC COUPLINGS ARE REQUIRED BETWEEN ALL DISSIMILAR METAL PIPING AND EQUIPMENT CONNECTIONS.	30. ALL BURIED PIPING SHALL BE BEDDED AND COVERED IN SAND, GRAVEL, OR CRUSHED STONE.
14. ISOLATE COPPER PIPE FROM HANGER OR SUPPORTS WITH ISOLATOR PAD (HAIR FELT LINING) SUPER STRUT MODEL C/15. FILL VOIDS BETWEEN PIPE AND WALL/FLOOR SLEEVES WITH FIRE-RATED FOAM SIMILAR TO CHASE TECHNOLOGY CORP. - C/C PR-855	31. AFTER COMPLETION OF PIPING TEST POTABLE WATER PIPING TO 125 LBS. PER SQ. INCH AND HOLD FOR 24-HOURS.
15. CONTRACTOR SHALL GUARANTEE ALL MATERIALS AND WORKMANSHIP FREE FROM DEFECTS FOR A PERIOD OF ONE (1) YEAR FROM DATE OF C.O. CORRECTION OF ANY DEFECTS SHALL BE COMPLETED WITHOUT ADDITIONAL CHARGE AND SHALL INCLUDE REPLACEMENT OR REPAIR OF ANY OTHER PHASE OF THE INSTALLATION WHICH MAY HAVE BEEN DAMAGED.	32. TEST DRAIN WASTE AND VENT PIPING BY FILLING TO LEVEL OF THE HIGHEST VENT.
16. PROVIDE 1/4" TRAP PRIMER LINE FOR ALL FLOOR DRAINS FROM THE NEAREST PLUMBING FIXTURE. PROVIDE MINIMUM 3" RADIUS, 1/4" PER FOOT SLOPE AROUND ALL FLOOR DRAINS.	33. AFTER INSTALLATION AND TESTING OF POTABLE WATER PIPING, STERILIZE ALL LINES IN ACCORD WITH CODES AND HEALTH DEPARTMENT REGULATIONS AND FLUSH AND FILL WITH CLEAN WATER.
17. PROVIDE ACCESS PANELS FOR ALL CONCEALED VALVES AND ALL WATER HAMMER ARRESTORS. ACCESS PANELS IN RATED WALLS MUST MAINTAIN THE RATING. ALL ACCESS PANELS MUST MATCH THE FINISH OF THE WALL IN WHICH IT IS INSTALLED.	34. FITCH POTABLE WATER LINES TOWARD DRAINS. INSTALL DRAIN WASTE AND VENT PIPING WITH MINIMUM SLOPE OF 1/4" PER FOOT FOR LINES UP TO 2 1/2" AND 1/8" PER FOOT FOR LINES 3" AND LARGER. PROVIDE A TWO INCH PIPE DIAMETER AIR GAP BETWEEN ALL INDIRECT WASTE AND THE RECOVERER. REFER TO ARCHITECTURAL DRAWINGS FOR LOCATIONS AND RATING OF FIRE WALLS AND FLOOR CEILING ASSEMBLIES.
	35. INSTALL FIRE STOP MATERIAL IN ACCORD WITH U.L. LISTING AT ALL PENETRATIONS.
	36. PIP WATER HEATER RELIEF VALVE TO EXTERIOR PER CODE OR TO FLOOR DRAIN.
	39. PROVIDE MAXVOLF SERIES OF 1" OR EQUAL FUEL GAS STRAINER PER NFPA 86 UPSTREAM OF SAFETY SHUTOFF VALVES. (PHONE NUMBER: (248) 356-1400).
	40. INSTALL WATER HEATERS IN ACCORDANCE WITH MANUFACTURERS INSTRUCTION AND ALL STATE AND LOCAL CODE REQUIREMENTS. WATER STORAGE SHALL HAVE A TEMPERATURE OF 140 DEGREES.
	41. ALL LAVATORIES AND HAND SINKS SHALL HAVE AN APPROVED ASSE 1016 OR ASSE 1017 DEVICE(S) FOR APPROVAL. COMPLY WITH ALL CODES.

RETURN AIR PLENUM NOTES
IF CEILING SPACE TO BE UTILIZED AS A RETURN AIR PLENUM, ALL MATERIALS LOCATED WITHIN THE PLENUM SHALL HAVE APPROVAL FOR SUCH USE. MATERIALS EXPOSED WITHIN PLENUMS SHALL BE NONCOMBUSTIBLE OR SHALL HAVE A FLAME SPREAD INDEX OF NOT MORE THAN 25 AND A SMOKE DEVELOPED INDEX OF NOT MORE THAN 50 WHEN TESTED IN ACCORDANCE WITH ASTM E 84 OR UL 723. FLORIDA MECHANICAL CODE SECTION 602, 2023
THE GENERAL CONTRACTOR SHALL INSURE THAT THE BUILDING ENVELOPE IS SUBSTANTIALLY AIR TIGHT. ALL ENVELOPE PENETRATIONS DUE TO PIPING, CONDUIT OR DUCT SHALL BE SEALED AIR TIGHT WITH A SEALANT APPROVED FOR USE IN A RETURN AIR PLENUM
PLUMBING CONTRACTOR SHALL INSULATE BOTH PRIMARY AND SECONDARY CONDENSATE DRAIN LINES WITH AN INSULATING MATERIAL APPROVED FOR USE WITHIN A RETURN AIR PLENUM
ALL REFRIGERATION PIPING INSULATIONS SHALL BE OF MATERIAL APPROVED FOR USE WITHIN A RETURN AIR PLENUM
ALL GAS PIPING SHALL BE OF GALVANIZED PIPING WITH SEALED FITTINGS AND SLEEVED IF REQUIRED BY STATE OR LOCAL CODES
ALL CEILING ELECTRICAL & FIRE ALARM MATERIALS SHALL BE RATED.



Elite Universal
 ARCHITECTURAL ENGINEERS
 3210 S. Ocoee Ave.
 Orlando, FL 32806
 407.325.3550
 www.eliteuniversalarchitects.com
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6001 Brick Court, Suite 201
 Miami, FL 33157
 (407) 756-6239
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CONSLTA T

**PLUMBING SPECIFICATIONS,
 ABBREVIATIONS, LEGEND
 AND NOTES**

MEDICAL OFFICE

SOUTHERN RAILWAY, BLDG #9 OAKLAND, FL 34787

REVISIONS:

#	DATE	DESCRIPTION
1	03/15/26	SUBMIT

N. A. S. AHMAD
 FLORIDA P.E. # 56095

DRAWN BY: AFS

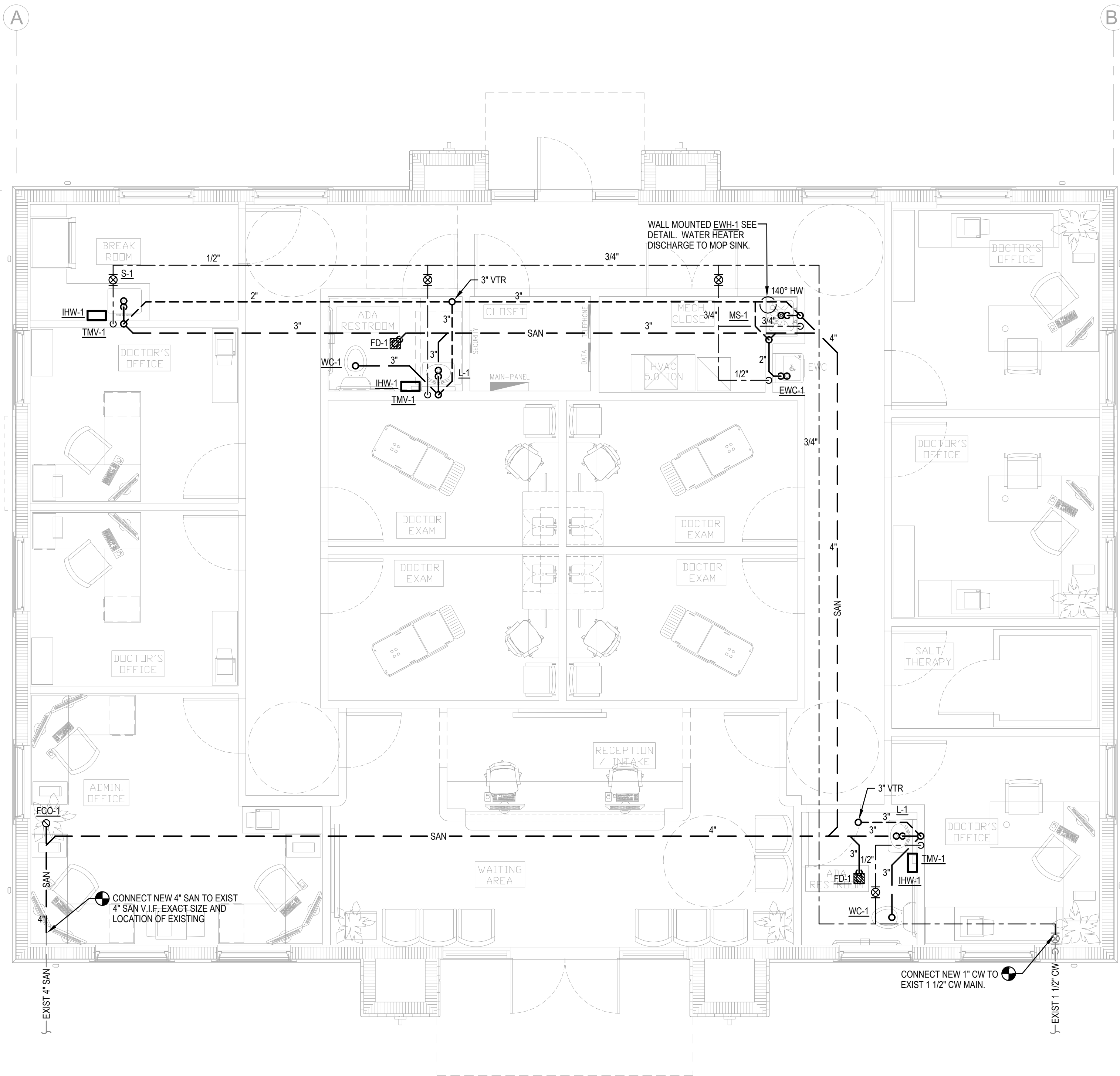
CHECKED BY: AFS

DATE: JANUARY 30, 2026

SHEET:

P000

Elite Universal Job No.
 90210



PLUMBING FIXTURE SCHEDULE

FIXTURE NO.	FIXTURE TYPE	WASTE CONN.	TRAP	VENT CONN.	C.W. CONN.	H.W. CONN.	MANUFACTURER / MODEL #	DESCRIPTION
WC-1	WATER CLOSET ADA TANK	3"	INTEG.	2"	1/2"	-	AMERICAN STANDARD 2998.012	RIM 16-1/2"AFF. TANK TYPE, VITREOUS CHINA, 1.6 GPF, SIPHON JET, ELONGATED BOWL, FLOOR MOUNTED, CLOSED COUPLED COMBINATION, BOLT CAPS. 'BRASSCRAFT' ANGLE STOP SUPPLY & ESCUTCHEON. 'BEMIS' SOLID PLASTIC, OPEN FRONT ELONGATED SEAT. FLUSH HANDLE TO BE ON OPEN SIDE OF TOILET CLEAR SPACE.
L-1	LAVATORY -- ADA TYPE 2 HANDLE FAUCET	1-1/2"	1-1/4"	1-1/2"	1/2"	1/2"	AMERICAN STANDARD 0355.012	LUCERNE 20" x 18" WALL HUNG, VITREOUS CHINA WITH INTEGRAL BACKSPASH. 'AMERICAN STANDARD' #2275.500, 4" CENTERS, C.P. METAL, TWO-LEVER HANDLE FAUCET. GRID STRAINER WITH TAILPIECE, C.P. BRASS P-TRAP WITH WALL ARM & ESCUTCHEONS. 'BRASSCRAFT' ANGLE VALVE SUPPLY RISERS W/LOOSE KEY STOPS. INSULATE P-TRAP AND SUPPLIES WITH 'TRUEBRO' INSULATION KIT. 'ZURN' 1231 CAST IRON FLOOR MOUNTED CARRIER FITTING WITH CONCEALED ARMS.
EW-1	ELECTRIC WATER COOLER	2"	1-1/2"	1-1/2"	1/2"		VRCGRN8WSK ADA	SINGLE WALL MOUNTED (ADA) STAINLESS STEEL BASIN BY ELKAY EZH2O® BOTTLE FILLING STATION & SINGLE HIGH EFFICIENCY, VANDAL-RESISTANT COOLER NON-FILTERED REFRIGERATED STAINLESS. CAPACITY TO DELIVER 8 GPH OF 50°F. WATER WITH INLET TEMP. OF 80°F. AND ROOM TEMP. OF 90°F. W/1/5 HP,120V, 60 CYCLE, 1 PH. W/ OVERLOAD PROTECTION 1-1/2" P-TRAP STOP VALVE, CORD & PLUG. ZURN #Z1225 CAST IRON FLOOR MOUNTED CARRIER FITTING WITH BEARING PLATE.
S-1	SINK SS SINGLE KITCHEN	2"	1-1/2"	1-1/2"	1/2"	1/2"	ELKAY LR2521	25" x 21" SINGLE COMPARTMENT SINK, 7-1/2" DEEP BOWL, 1 HOLE, #18 GAUGE, TYPE 304 NICKEL BEARING STAINLESS STEEL, SELF RIMMING, 'CHICAGO' #2300-CP, C.P. SINGLE LEVER HANDLE FAUCET. C.P. STRAINER WITH TAILPIECE, C.P. BRASS P-TRAP WITH WALL ARM & ESCUTCHEONS. 'BRASSCRAFT' ANGLE VALVE SUPPLY RISERS W/LOOSE KEY STOPS.
MS-1	MOP SINK / JANITOR SINK	3"	3"	1-1/2"	3/4"	3/4"	ELKAY 000	WALL MOUNTED #14 GAUGE STAINLESS STEEL SERVICE SINK WITH VERTICAL AND HORIZONTAL COVERED CORNERS AND ROLLED RIMS. FULL LENGTH 12"HIGH BACK-SLASH. FURNISHED WITH HANGER AND STAINLESS STEEL SUPPORT BRACKETS. SUPPLY WITH LK405 3/4" HOSE THREAD SPOUT AND LK173 CAST IRON "P" TRAP STANDARD.

NOTE: ALL PLUMBING FIXTURES SHALL BE SUPPLIED WITH WATTS "FLOODSAFE" AUTO-SHUT OFF CONNECTOR TYPE SUPPLY RISERS WITH BRAIDED STAINLESS STEEL CONNECTORS. THIS SHALL INCLUDE ALL FAUCETS, TOILETS, WASHING MACHINES, DISHWASHERS & ICE MAKERS. PROVIDE FULLY ACCESSIBLE SHOCK ARRESTORS ON SUPPLIES TO WASHING MACHINES, ICE MAKERS AND DISHWASHERS.

PLUMBING SPECIALTIES SCHEDULE

FIXTURE NO.	FIXTURE TYPE	WASTE CONN.	TRAP	VENT CONN.	C.W. CONN.	H.W. CONN.	MANUFACTURER / MODEL #	DESCRIPTION
ECO-1	EXTERIOR CLEANOUT	SEE PLAN	-	-	-	-	ZURN CO-2413	PVC CLEAN-OUT WITH COUNTER SUNK PLUG WITH PVC 2-WAY CLEAN-OUT TEE, (LAWN, IN FLUSH 12"x12"x4" CONCRETE PAD.)
FCO-1	FINISHED FLOOR CLEANOUT	SEE PLAN	-	-	-	-	ZURN Z1400	"LEVEL-TROL" ADJUSTABLE FINISHED FLOOR CLEANOUT, DURA-COATED CAST IRON BODY WITH GAS AND WATERTIGHT ABS TAPERED THREADED PLUG, AND ROUND SCORIATED SECURED TOP. ADJUST TO FINISHED FLOOR (COORDINATE TOP FINISH WITH ARCHITECT).
FD-1	FLOOR DRAIN STANDARD ROUND	SEE PLAN	-	-	1/2"	-	ZURN ZN415B	DURA-COATED CAST IRON BODY WITH BOTTOM OUTLET, COMBINATION INVERTIBLE MEMBRANE CLAMP AND ADJUSTABLE COLLAR WITH SEEPAGE SLOTS AND TYPE 'B' POLISHED NICKEL BRONZE STRAINER. PROVIDE 1/2" COLD WATER TRAP PRIMER CONNECTION
SA	SHOCK ABSORBER	-	-	-	-	-	SIoux CHIEF 650 SERIES	SIZE PER MANUFACTURER'S RECOMMENDATIONS. PDI CERTIFIED. WHA-1: SIZE "A" WHA-2: SIZE "B" REFER TO RISER DIAGRAMS FOR ARRESTOR LOCATIONS.
TPV-1	TRAP PRIMER VALVE	-	-	-	1/2"	-	PRECISION PLUMBING PRODUCTS PR-500	CORROSION RESISTANT BRASS TRAP PRIMER, 1/2" COLD WATER INLET AUTOMATICALLY OPERATED BY PRESSURE DROP OF 10 PSIG.
TMV-1	THERMOSTATIC MIXING VALVE	-	-	-	3/8"	3/8"	LEONARD 170 LF	POINT OF USE MIXING VALVE. 3/8" INLETS, 3/8" OUTLETS, COMPRESSION CONNECTIONS, 0.25 GPM MINIMUM FLOW RATE AND 125 PSI MAX. PRESSURE. PROVIDE WITH BP-COLD WATER BY - PASS, CP - CHROME PLATED BRKT - MOUNTING BRACKET. FACTORY ASSEMBLED AND TESTED.
WCO-1	WALL CLEAN-OUT	SEE PLAN	-	-	-	-	ZURN Z1446	CAST IRON CLEAN-OUT TEE, DURA COATED BODY, GAS AND WATERTIGHT ABS TAPERED THREAD PLUG AND ROUND, SMOOTH STAINLESS STEEL WALL ACCESS COVER WITH SECURING SCREW.

ALL PLUMBING FIXTURES SHALL BE SUPPLIED WITH WATTS "FLOODSAFE" AUTO-SHUT OFF CONNECTOR TYPE SUPPLY RISERS WITH BRAIDED STAINLESS STEEL CONNECTORS. THIS SHALL INCLUDE ALL FAUCETS, TOILETS, WASHING MACHINES, DISHWASHERS & ICE MAKERS. PROVIDE FULLY ACCESSIBLE SHOCK ARRESTORS ON SUPPLIES TO WASHING MACHINES, ICE MAKERS AND DISHWASHERS.

JANITOR SINK ELECTRIC WATER HEATER SCHEDULE

UNIT #	GALLONS	MANUFACTURER		KW INPUT	GPH REC. @ 60°F RISE	VOLTAGE	DRY WEIGHT (LBS.)	DIMENSIONS	ENERGY FACTOR
		MANUFACTURER	MODEL #						
EW-1	20	AO SMITH	DEL-20	1.5	10	120V-1PH	54	18 1/4" H x 18" Ø	0.92

PROVIDE WITH THERMAL EXPANSION TANK (AMTROL THERM-X-TROL ST-12 OR EQUAL) EXPANSION TANK SUPPORT OR THERMAL EXPANSION VALVE, T & P RELIEF VALVE AND WATER HEATER DRAIN PAN. INSTALL PER SECTION 415.5 PLUMBING CODE.

ELECTRIC INSTANTANEOUS WATER HEATER SCHEDULE

UNIT #	LOCATION	TANKLESS	MANUFACTURER		KW INPUT	GPH REC. @ 90°F RISE	VOLTAGE	DRY WEIGHT (LBS.)	DIMENSIONS	ENERGY FACTOR
			MANUFACTURER	MODEL #						
IWH-1	SEE DRAWINGS	INSTANTANEOUS	CHRONOMITE	SR-15L/120	1.8		120V	5.00	H 6-1/4" (H) x W 9-5/8" x D 2-3/4"	0.99

INSTALLATION & OPERATION NOTES: TURN THE HOT WATER FIXTURE TO ACTIVATE THE FLOW SWITCH. THE FLOW SWITCH ACTIVATES AT 0.65 GALLONS PER MINUTE (GPM) AND DEACTIVATES AT 0.50 GALLONS PER MINUTE (GPM). IF YOU INCREASE THE FLOW RATE OF THE WATER ABOVE THE KW CAPACITY, THE WATER TEMPERATURE WILL BECOME COOLER. COLD WATER CAN ALWAYS BE MIXED JUST AS WITH A CONVENTIONAL SYSTEM IF USING A TWO HANDLE FAUCET. SELECT THE APPROPRIATE FLOW CONTROL. ONCE YOU DECREASE THE HOT WATER FLOW RATE TO 0.50 GPM THE UNIT WILL DEACTIVATE. IT IS NOT ADVISABLE TO OPERATE THE HEATER AT 0.50 GPM HOT WATER FLOW RATE. SET WATER HEATER TEMPERATURE AT 110°F MAX. INSTALL PER SECTION 415.5 PLUMBING CODE, WATER TEMPERATURE LIMITING DEVICE THAT CONFORMS TO ASSE 1070.

SLEEVES:

UNDERGROUND, EXTERIOR-WALL PIPE PENETRATIONS:
INSTALL CAST IRON "WALL PIPES" FOR SLEEVES. CAST IRON: CAST OR FABRICATED "WALL PIPE" EQUIVALENT TO DUCTILE IRON PRESSURE PIPE, WITH PLAIN ENDS AND INTEGRAL WATER STOP, UNLESS OTHERWISE INDICATED (USE PVC AS INDICATED). SEAL PIPE PENETRATIONS USING MECHANICAL SLEEVE SEALS. SELECT SEAL SIZE TO ALLOW FOR 1-INCH (25 MM) ANNULAR CLEAR SPACE BETWEEN PIPE AND SLEEVE FOR INSTALLING MECHANICAL SLEEVE SEALS.

MECHANICAL SLEEVE SEAL INSTALLATION:
SELECT TYPE AND NUMBER OF SEALING ELEMENTS REQUIRED FOR PIPE MATERIAL AND SIZE. POSITION PIPE IN CENTER OF SLEEVE. ASSEMBLE MECHANICAL SLEEVE SEALS AND INSTALL IN ANNULAR SPACE BETWEEN PIPE AND SLEEVE. TIGHTEN BOLTS AGAINST PRESSURE PLATES THAT CAUSE SEALING ELEMENTS TO EXPAND AND MAKE WATER TIGHT

MECHANICAL SLEEVE SEALS
A. DESCRIPTION: MODULAR SEALING ELEMENT UNIT, DESIGNED FOR FIELD ASSEMBLY, TO FILL ANNULAR SPACE BETWEEN PIPE AND SLEEVE.

- SEALING ELEMENTS: EPDM OR NBR INTERLOCKING LINKS SHAPED TO FIT SURFACE OF PIPE. INCLUDETYPE AND NUMBER REQUIRED FOR PIPE MATERIAL AND SIZE OF PIPE.
- PRESSURE PLATES: STAINLESS STEEL. INCLUDE TWO FOR EACH SEALING ELEMENT.
- CONNECTING BOLTS AND NUTS: STAINLESS STEEL OF LENGTH REQUIRED TO SECURE PRESSURE PLATES TO SEALING ELEMENTS. INCLUDE ONE FOR EACH SEALING ELEMENT.

